

Item2item service

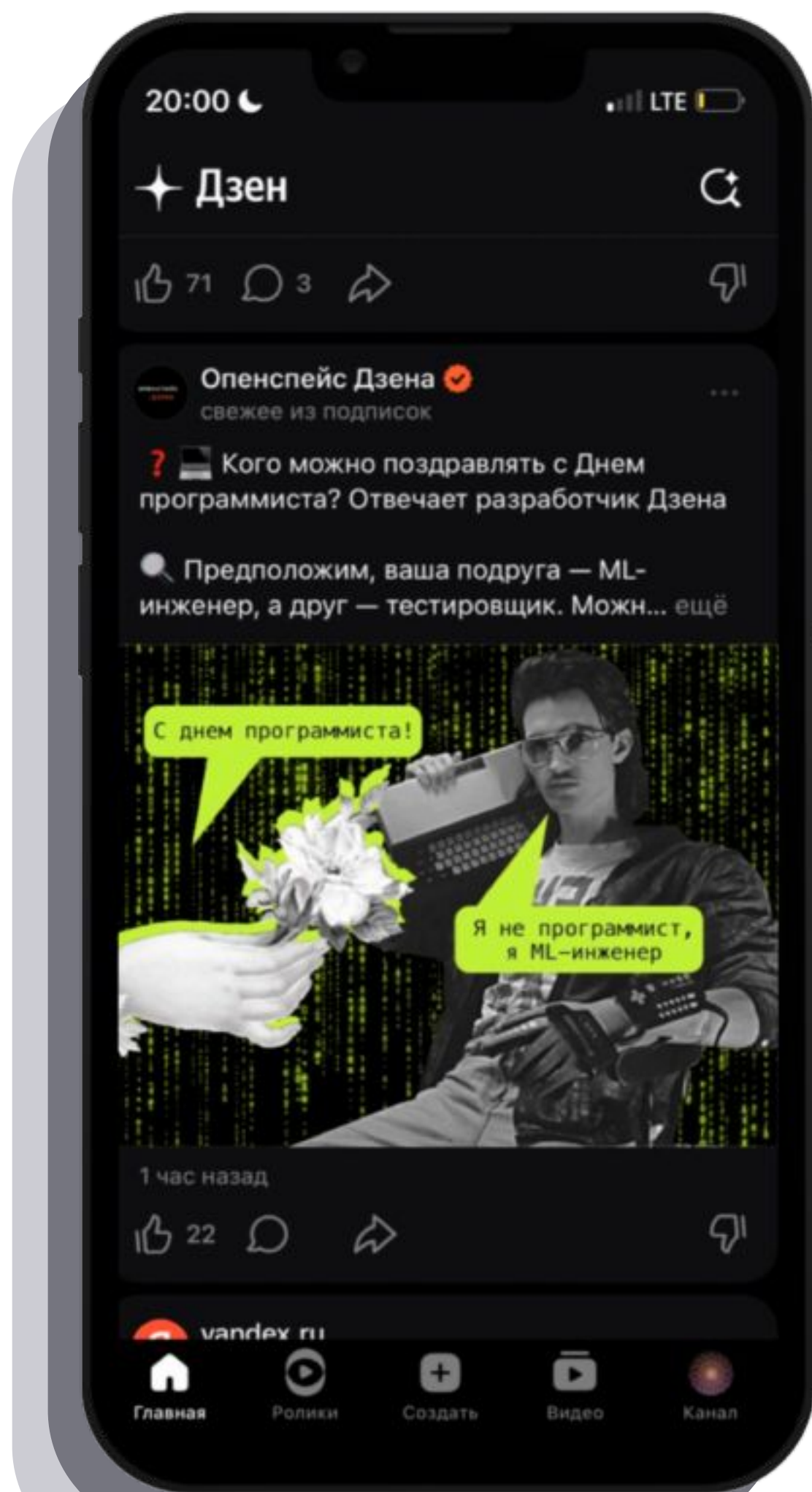
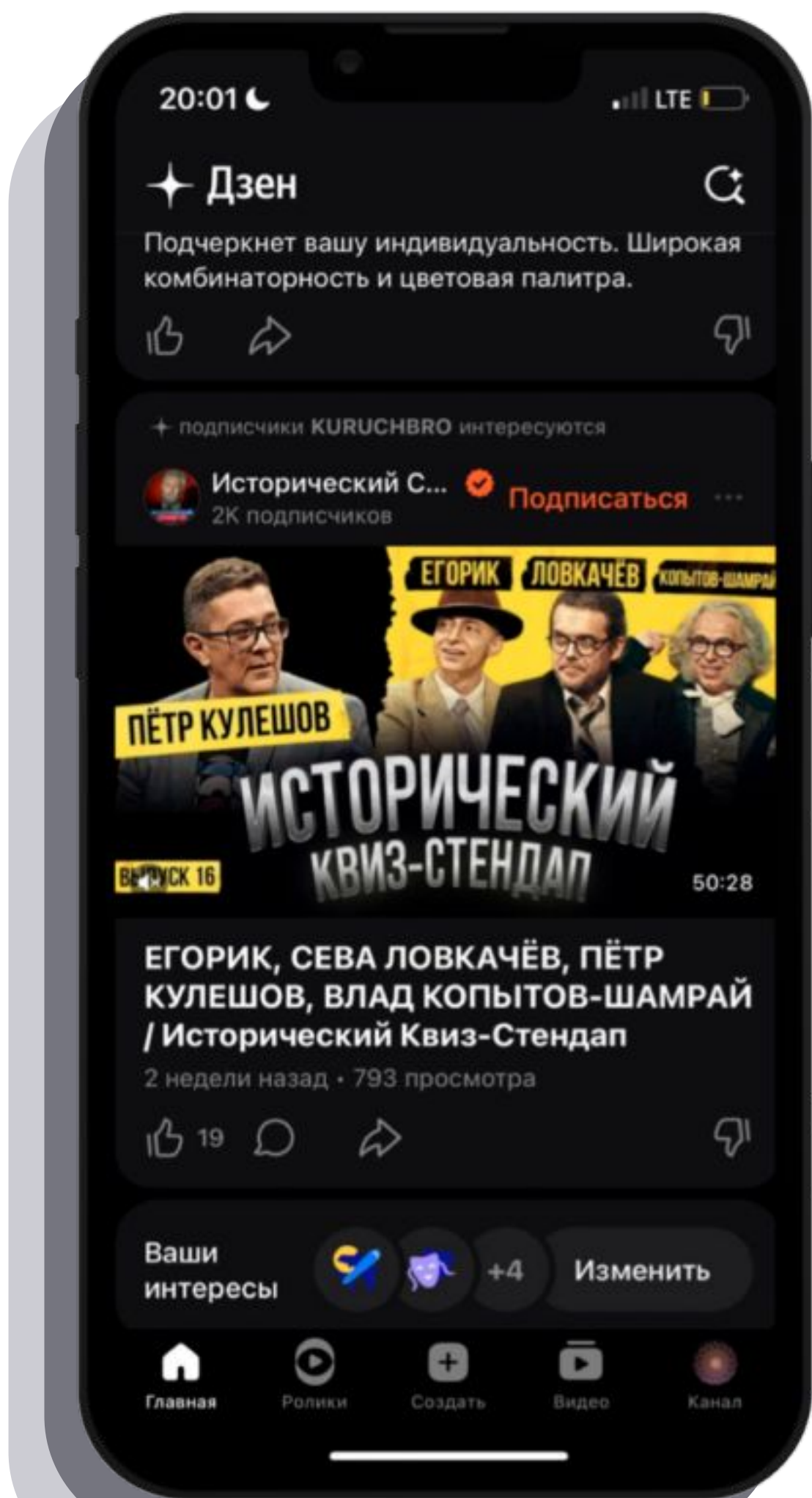
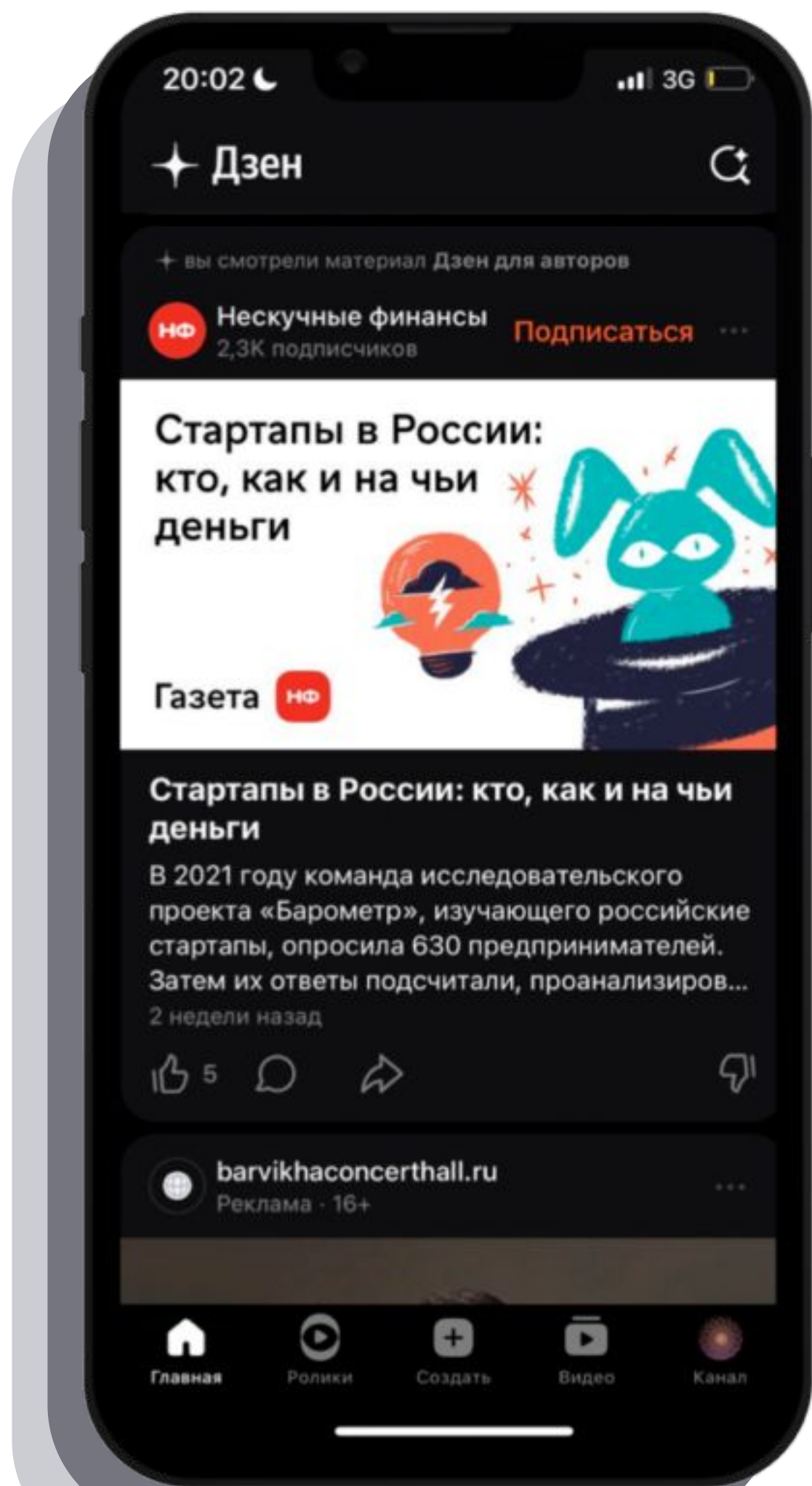
Vladislav Goncharenko

Senior lecturer,
ML team lead at Dzen



MIPT, spring 2026

Dzen



Dzen in numbers

DAU 30M+ users

MAU 70M+ users

35 mins a day

10K RPS for recommendations

100M+ users with history

5 mln candidate items

10 bills events a day

500ms for response

Plan

- ✦ Metrics in Dzen
- ✦ item2item recommendations
- ✦ Dzen recommendations and i2i
- ✦ Results

Dzen metrics and casualties found



Top-level service metrics

DAU

NPS

Timespent

Top-level service metrics

How to grow DAU?

- ✦ Marketing campaigns
- ✦ Exclusive content
- ✦ SEO



Дзен

Saint
HighLoad++
2023
HL

Top-level service metrics

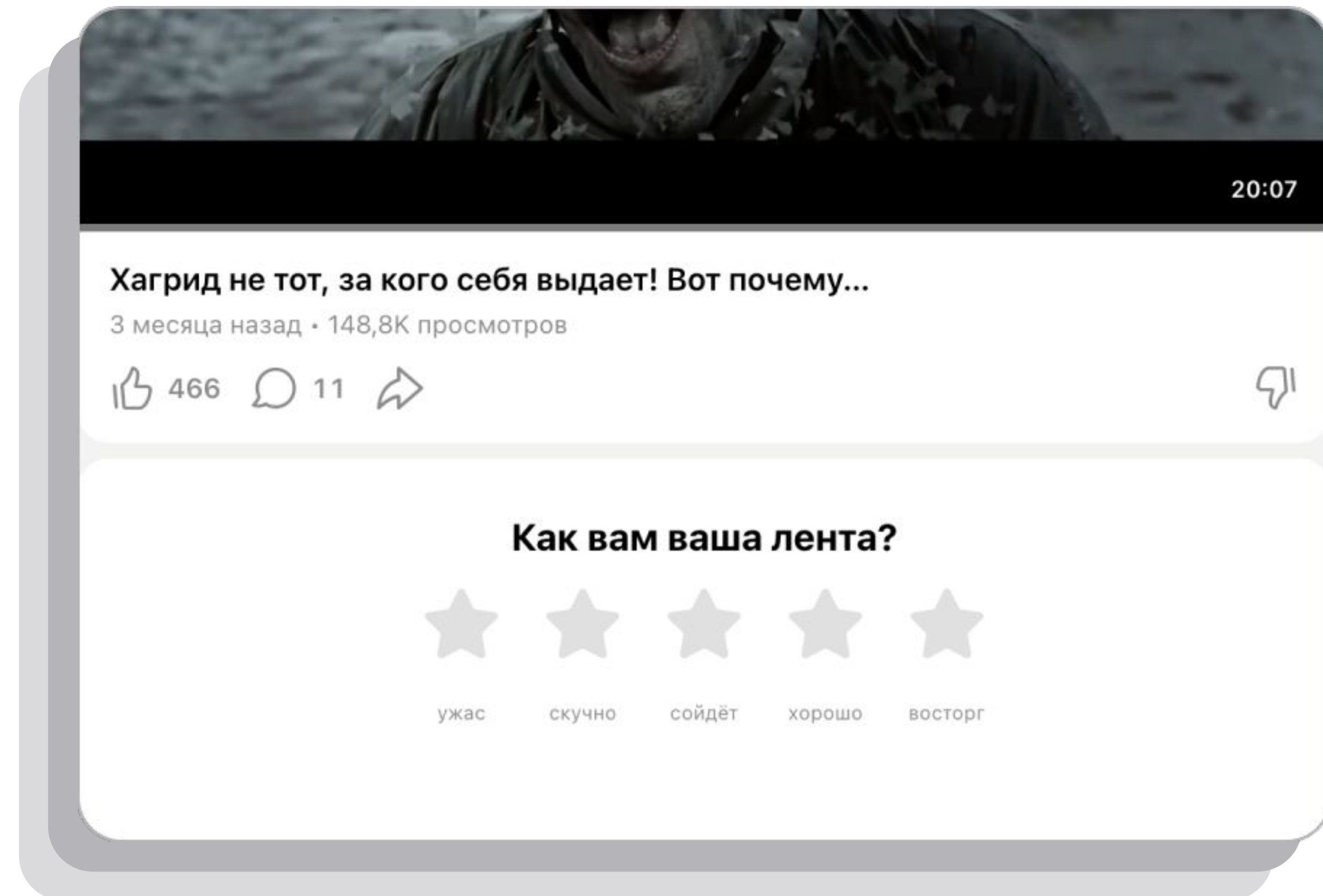
How to grow Timespent?

- ✦ Increase predictions quality
 - ✦ Interpret signal better
 - ✦ Try new (more complex) models
- ✦ Grow content base

Top-level service metrics

How to grow Net Promoter Score (NPS)?

- ✦ Conduct survey:
ask users



Top-level service metrics

How to grow Net Promoter Score (NPS)?

- ✦ Conduct survey:
ask users

The image shows a screenshot of a survey interface from the Dzen mobile application. At the top, there is a video player with the title "ХАГРИД - РОЖИРАТЕЛЬ СМЕРТИ" and a timestamp of 19:08. Below the video, the text "Хагрид не тот, за кого себя выдает! Вот почему..." is displayed, along with "3 месяца назад • 148,8К просмотров". Below this, there are icons for likes (466), comments (11), and shares. The main part of the survey is titled "Что не так с вашей лентой?" and contains four checkboxes: "нет интересных авторов", "однообразная лента", "слишком много рекламы", and "некачественный контент". At the bottom, there is a text input field with the placeholder "Вы там в своём уме?" and an orange button labeled "Отправить обратную связь".

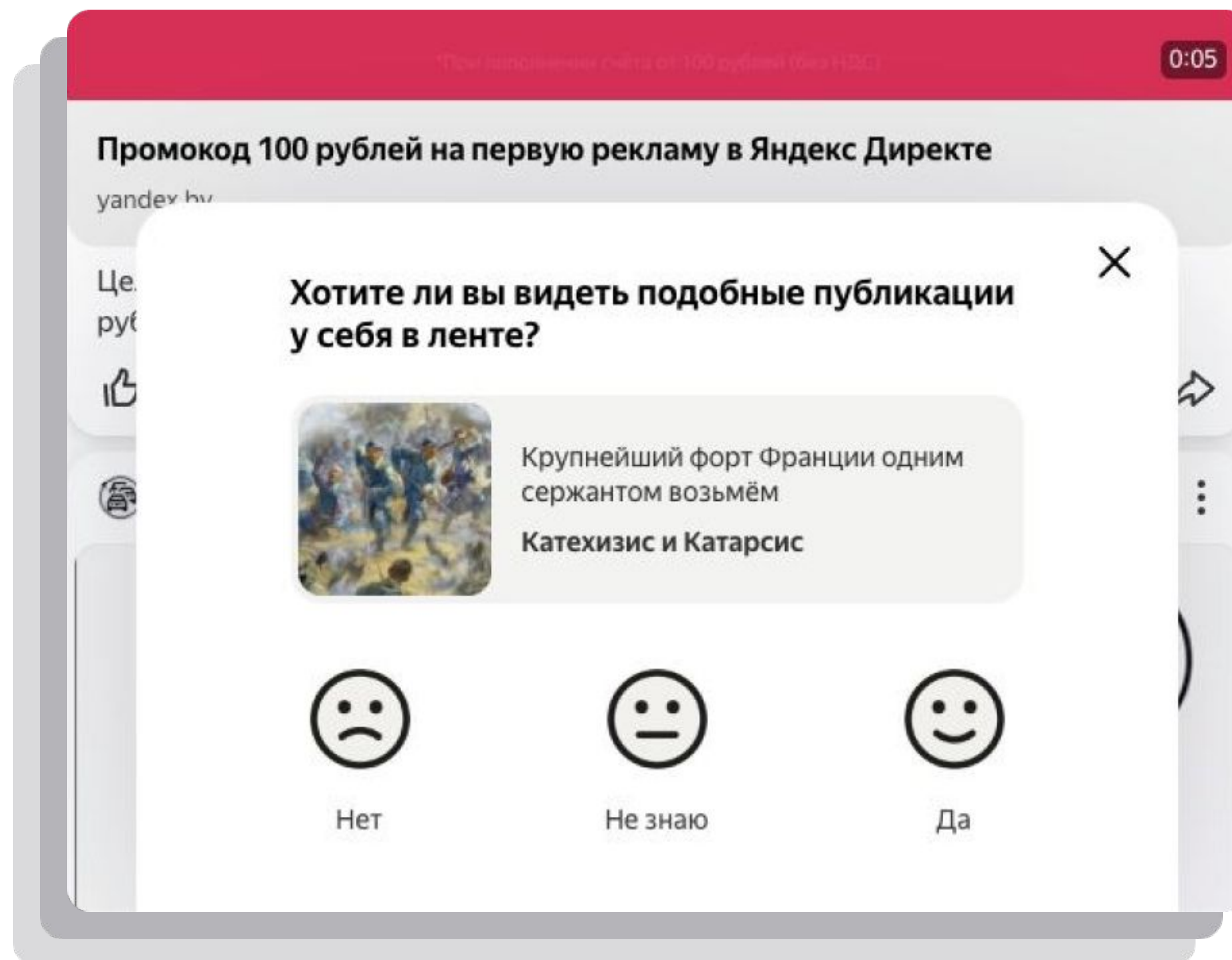
Top-level service metrics

NPS, survey results:

- ✦ Discontent of some recommendations:
 - ✦ «Don't understand why do you show this, I haven't liked anything similar»
- ✦ Not all interests of a user are presented
 - ✦ «Sometimes I read about motorsport, but I don't see it in feed»
- ✦ Advertising pressure
- ✦ Bugs
- ✦ Weater
- ✦ etc

What do we want?

Grow sensitive proxy metric
for NPS - CSI of
recommendations



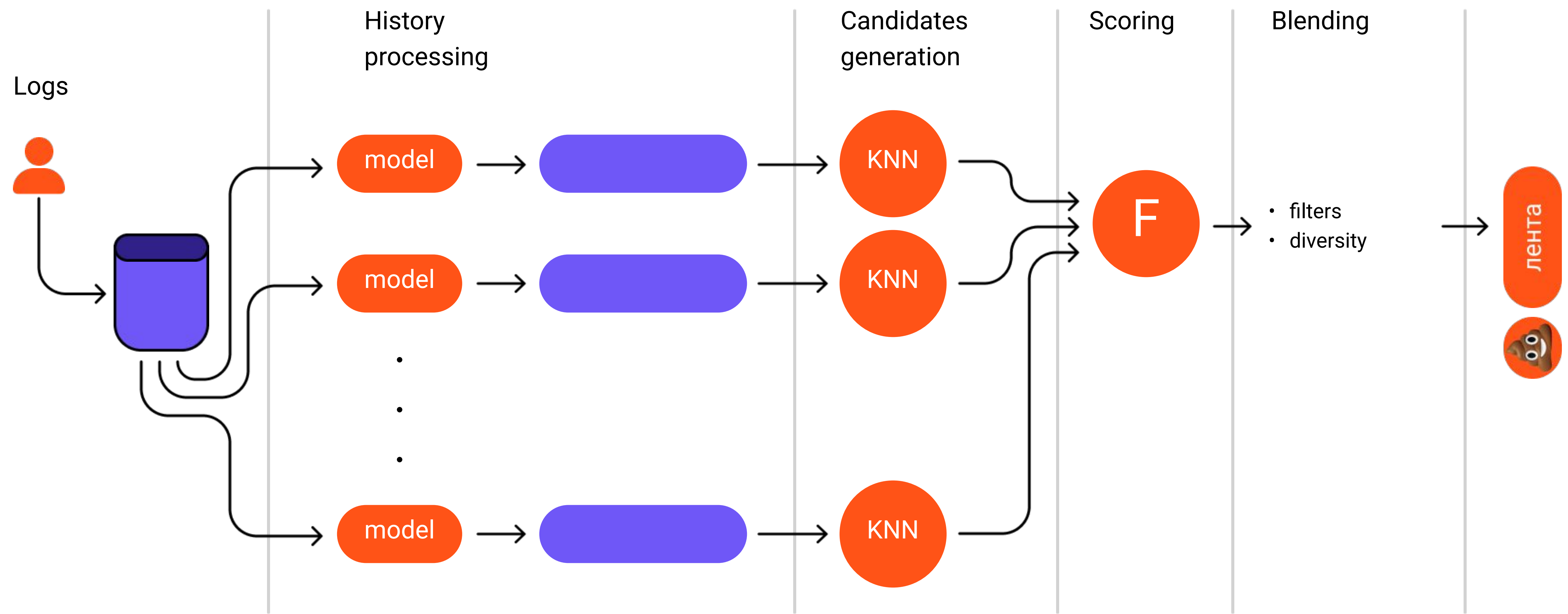
Let's debug

«Don't understand why do you show this, I haven't liked anything similar»

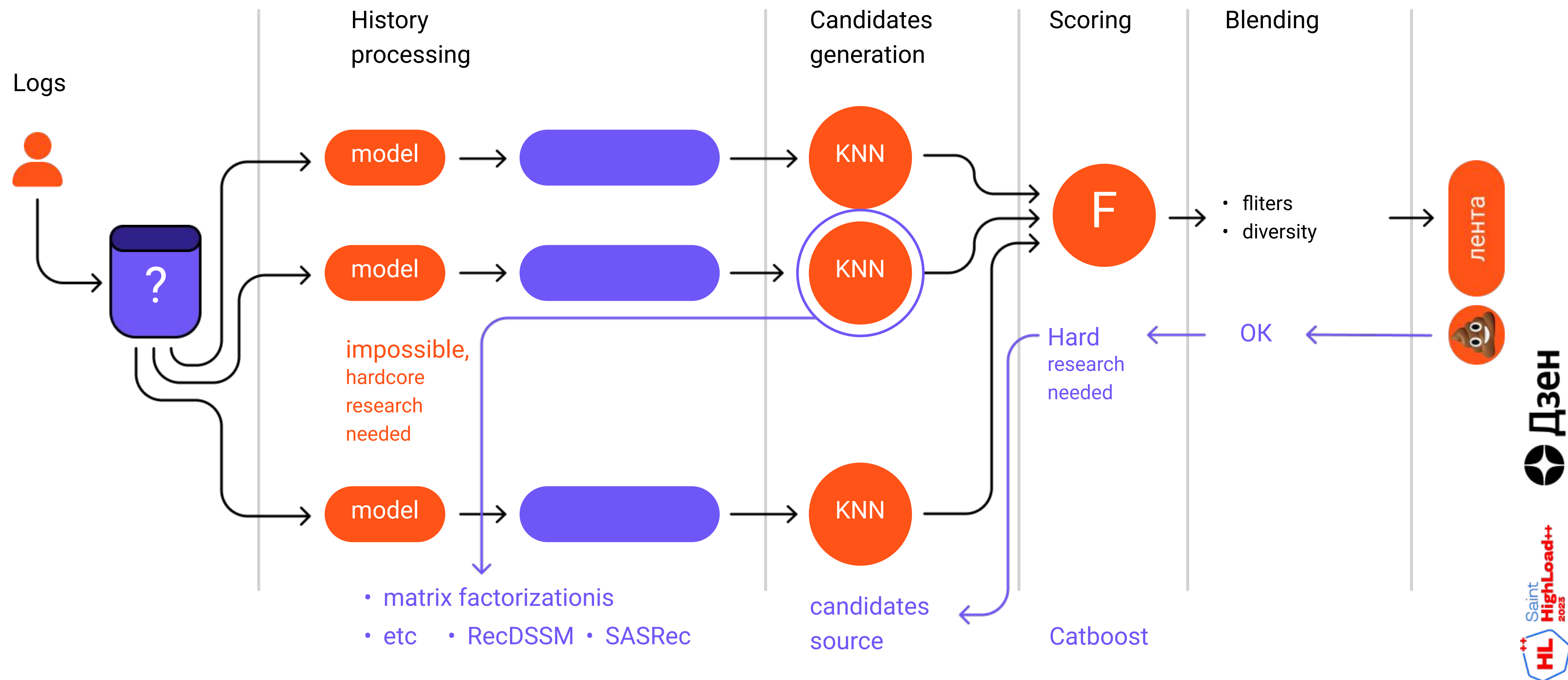
(c) financial director



Recommendation system overview



Feed debug

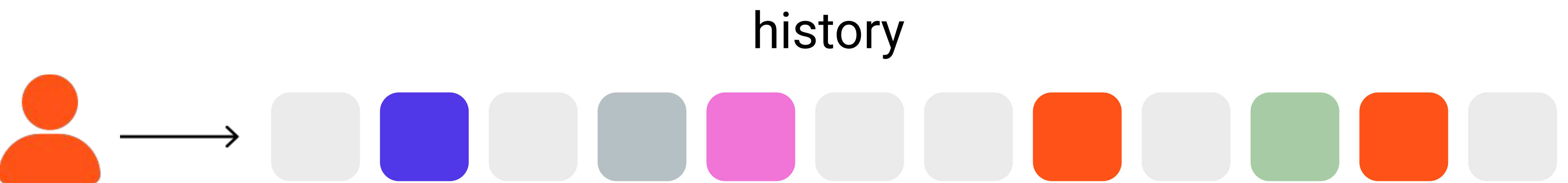


What do we want?

Make candidates generation
process **simple**

Problems

Some items are not similar to user history



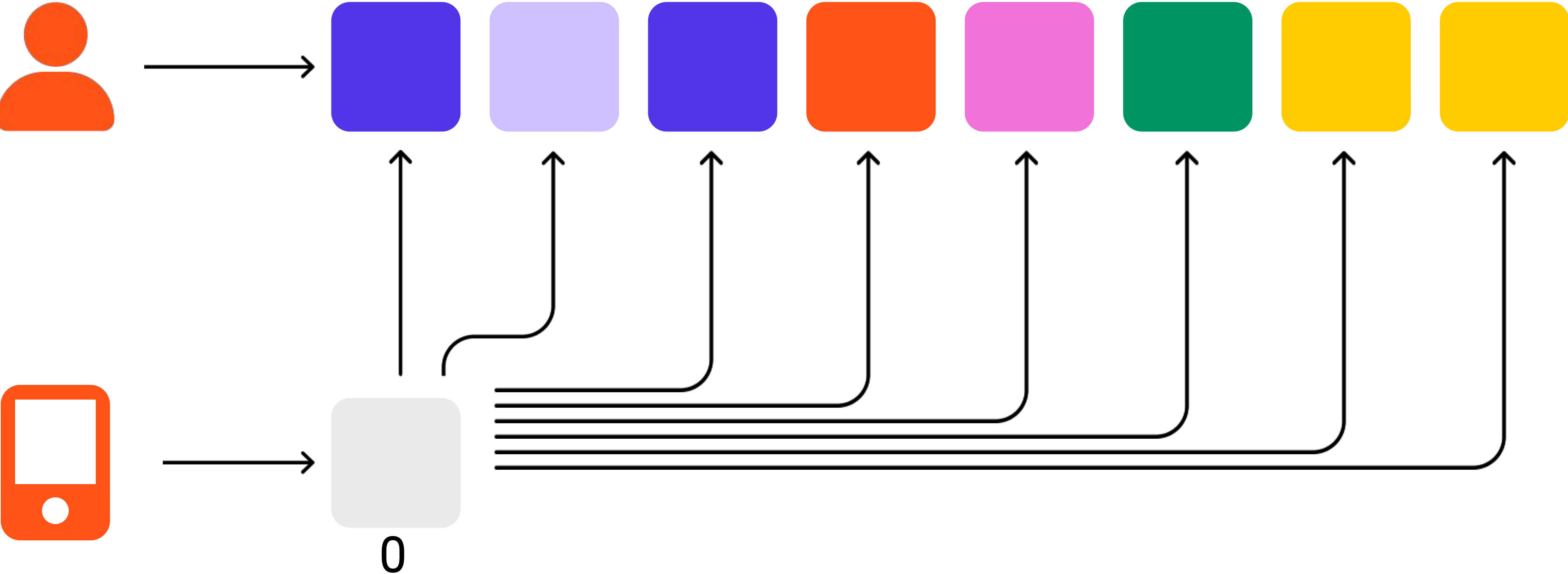
Positives

- ✦ likes
- ✦ long watches
- ✦ readups
- ✦ ...

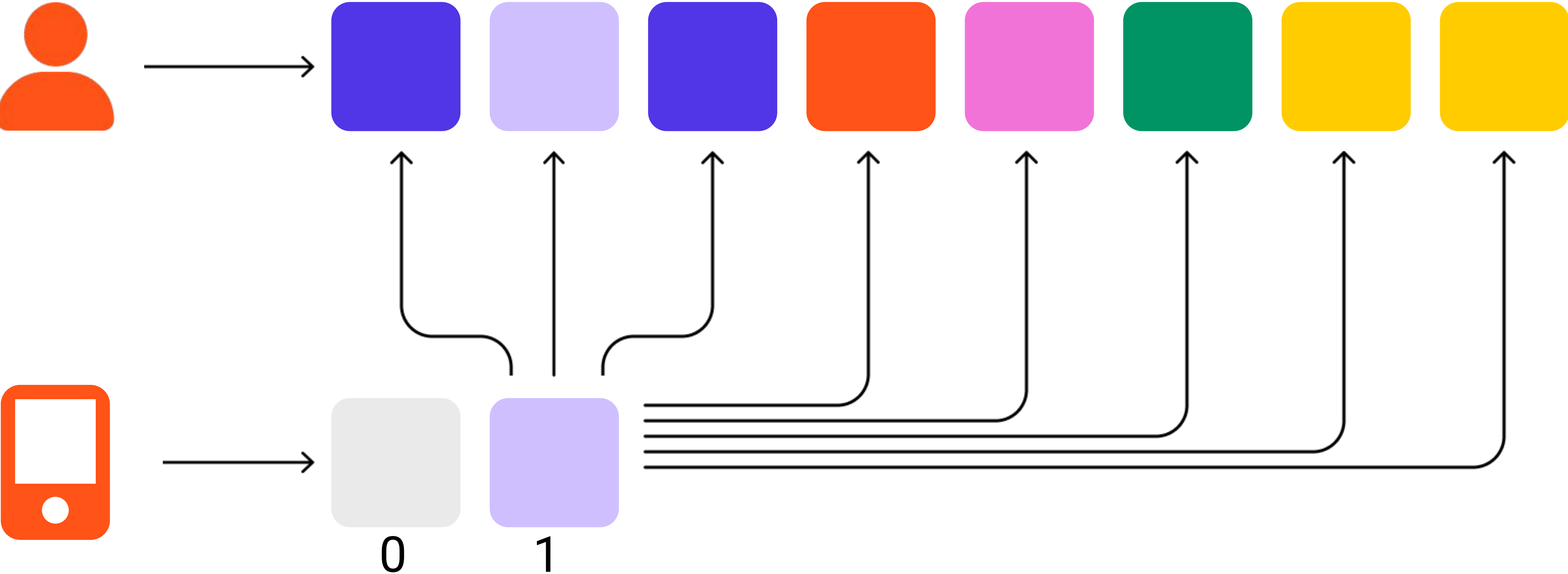
Problems



Problems

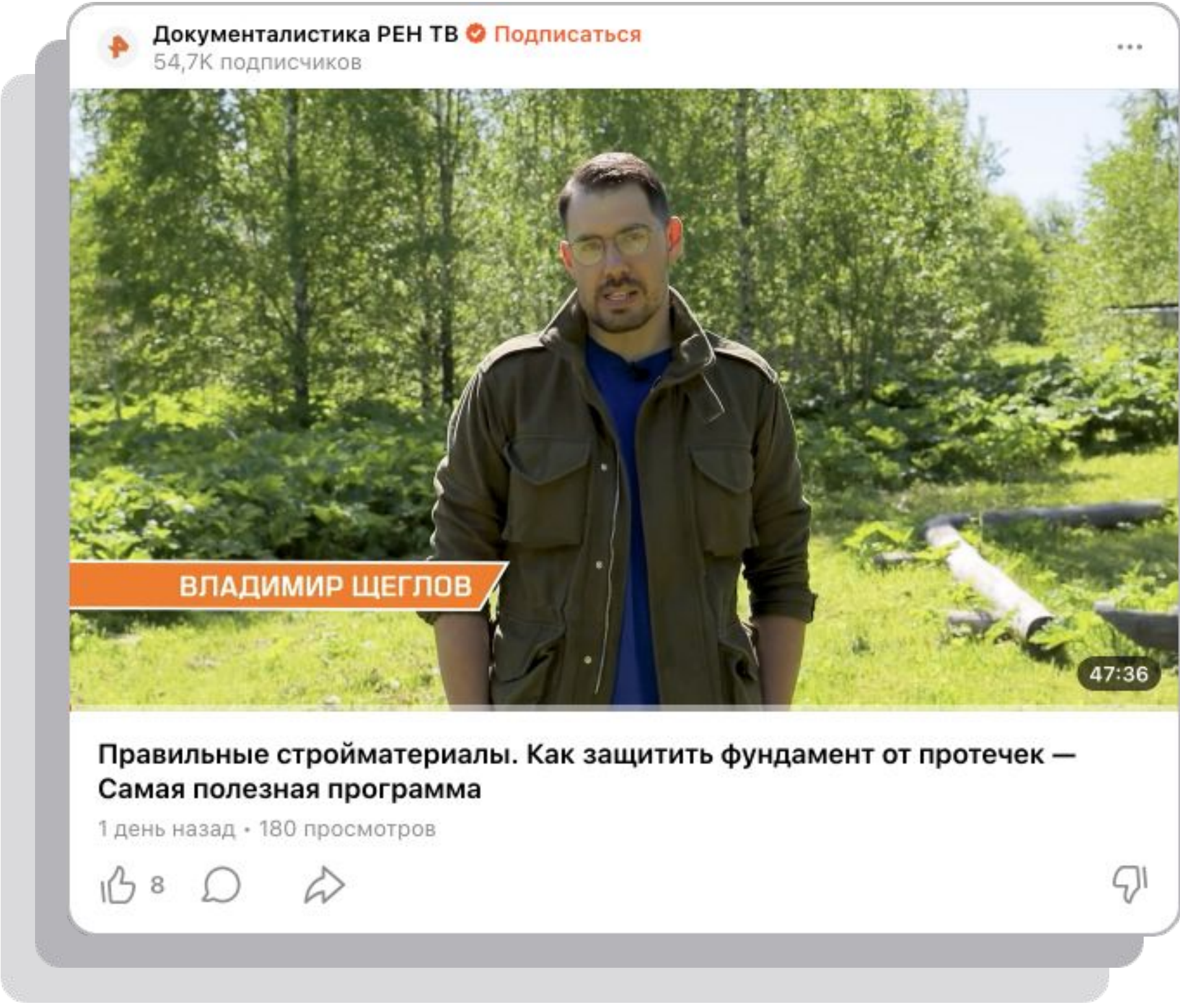
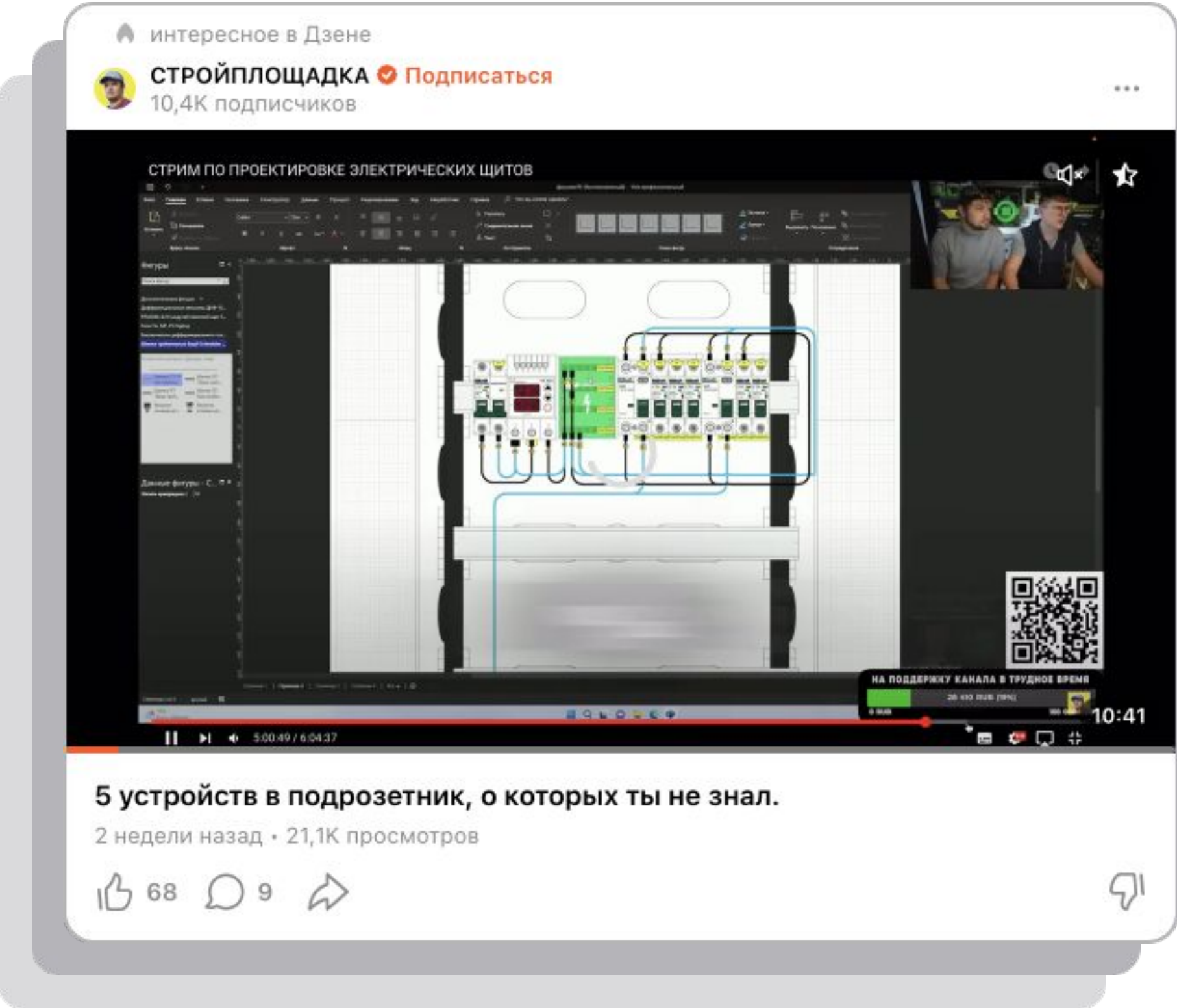


Problems



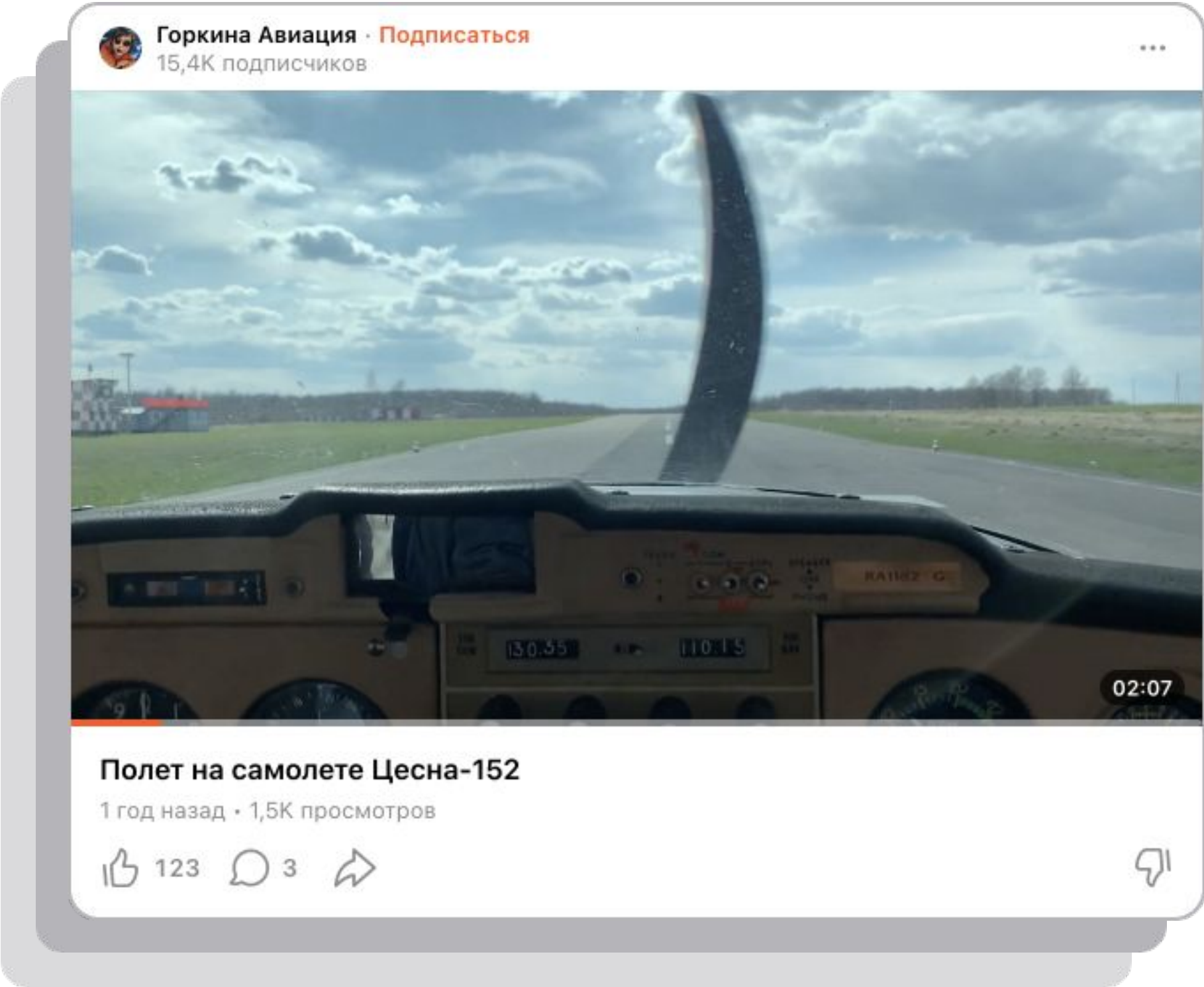
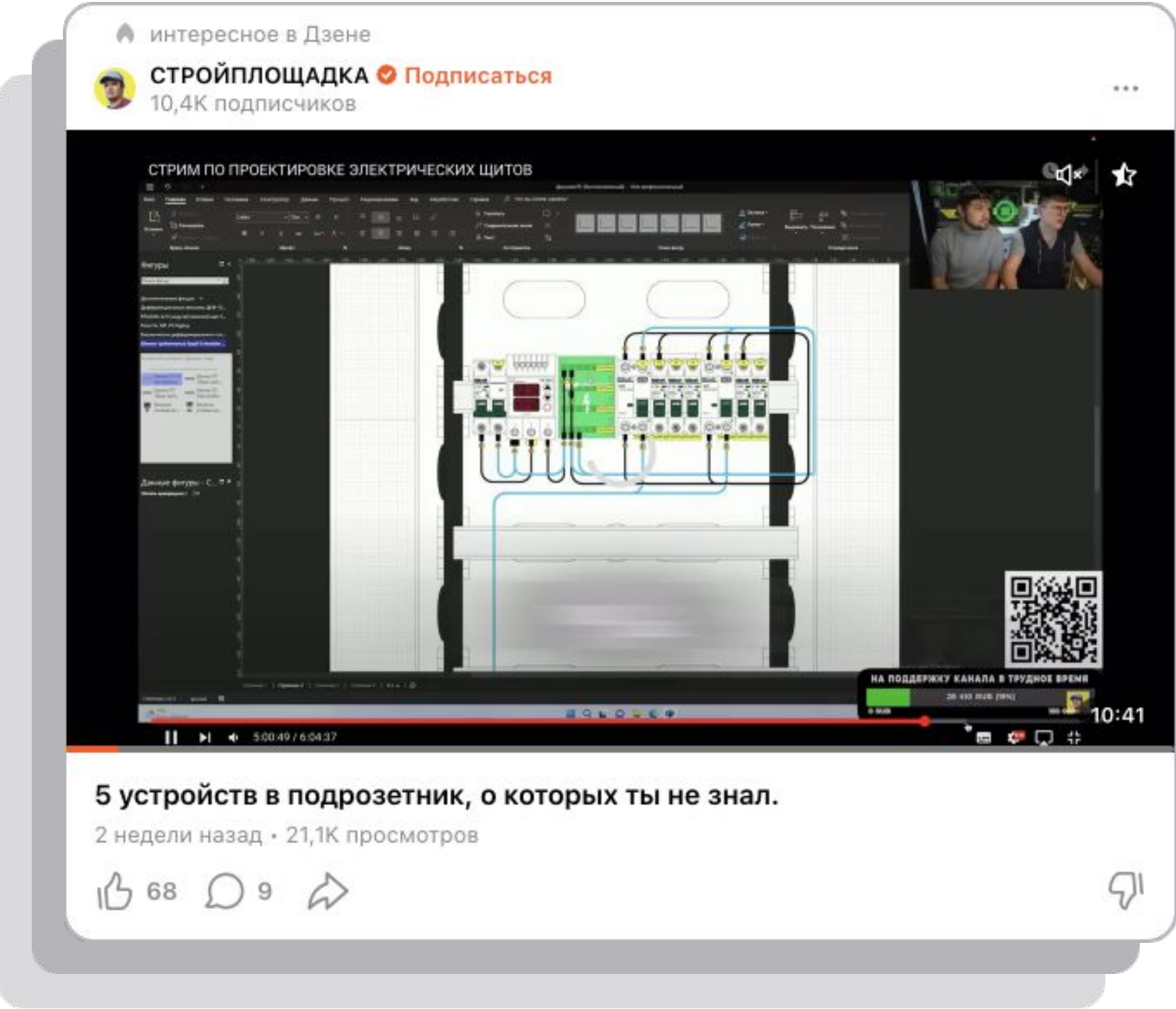
How to know if two items are similar?

Crowd source labeling with Toloka



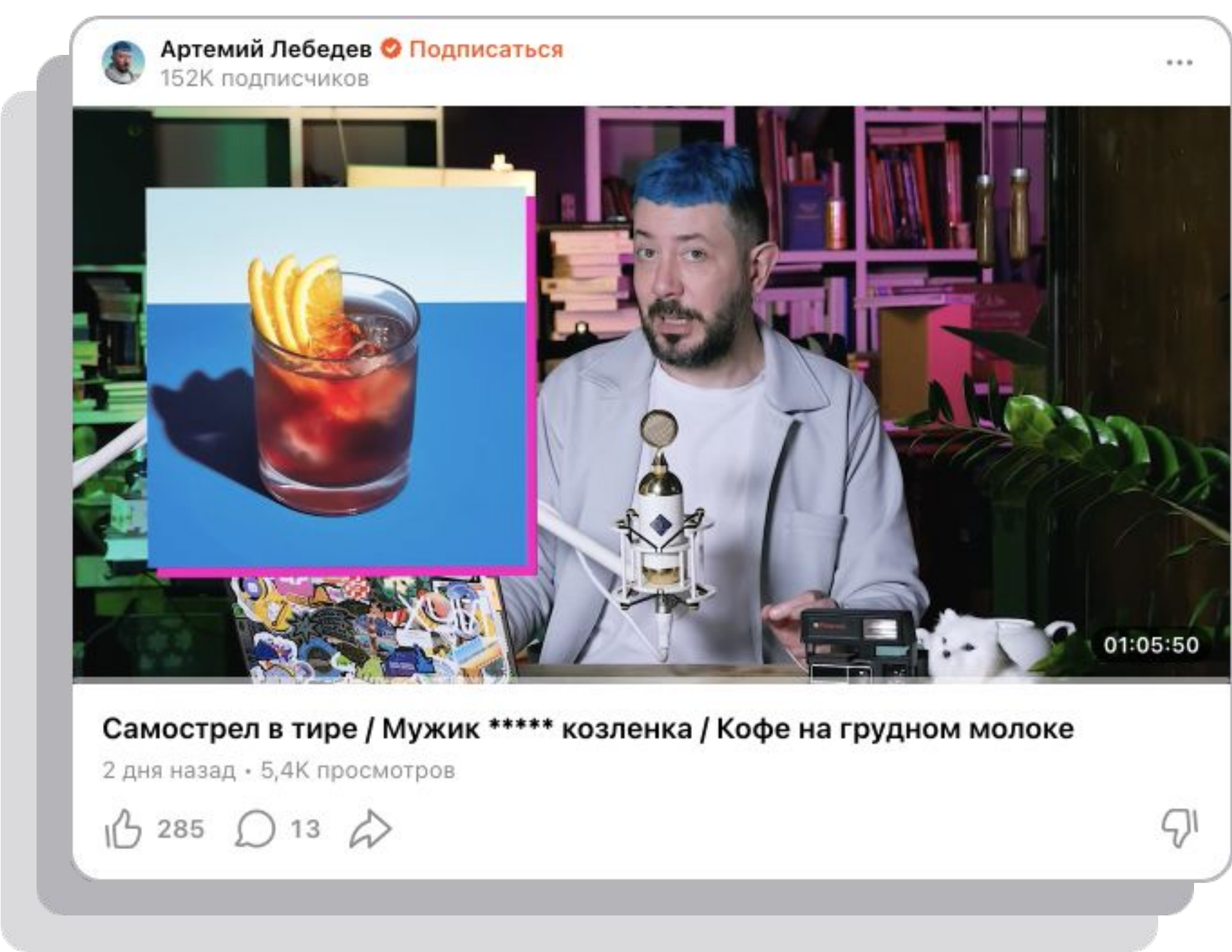
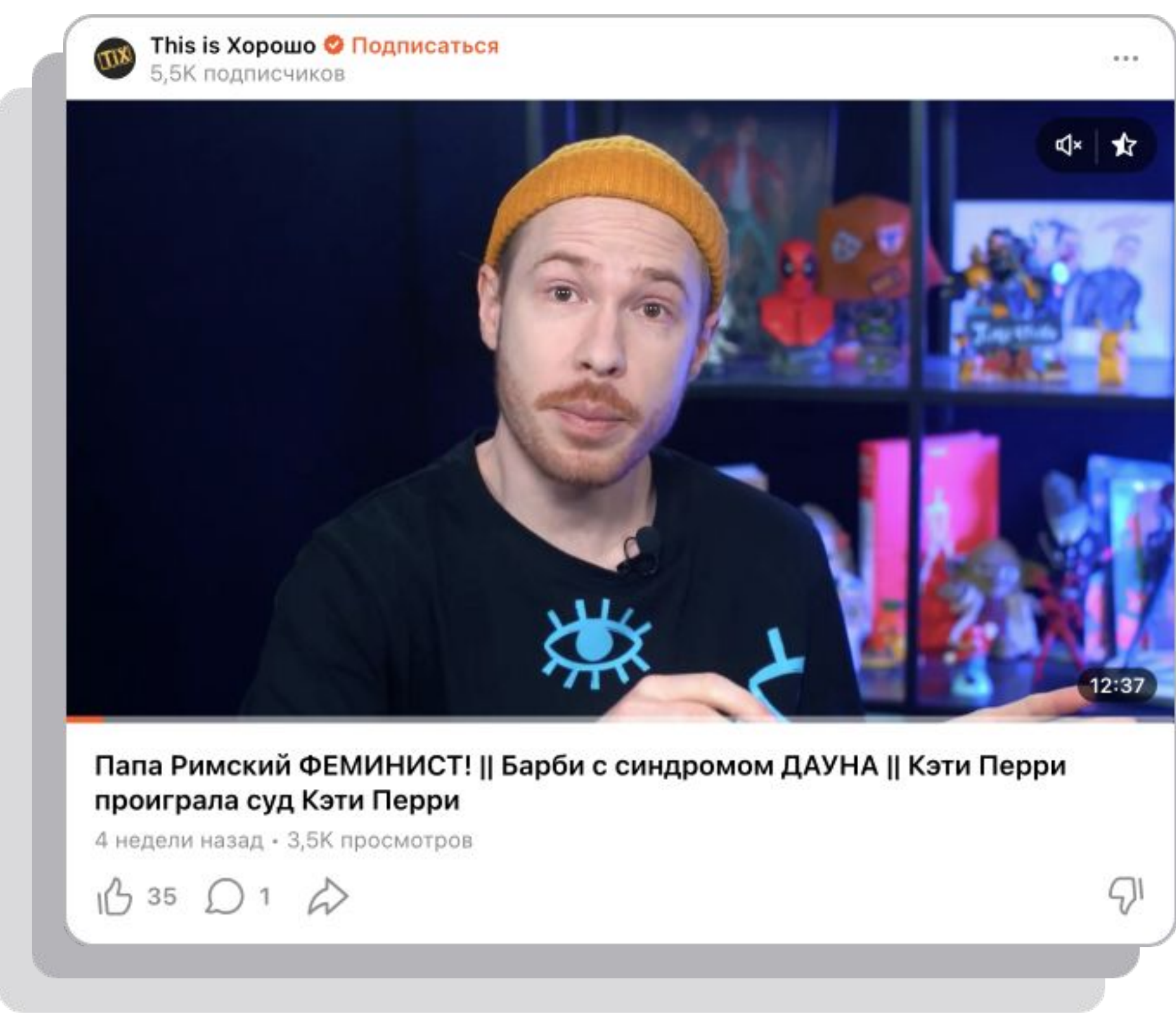
How to know if two items are similar?

Crowd source labeling with Toloka



How to know if two items are similar?

Crowd source labeling with Toloka



How to know if two items are similar?

Crowd source labeling with Toloka

This is Хорошо

✓ Подписаться

5,5K подписчиков

Папа Римский ФЕМИНИСТ! || Барби с синдромом ДАУНА || Кэти Перри проиграла суд Кэти Перри

4 недели назад • 3,5K просмотров

35

1

BigJapan125

Подписаться

586 подписчиков

Обзор нового и технологичного Geely Monjaro 🤖 Какой ты Лухари из Китая??

3 месяца назад • 30,5K просмотров

78

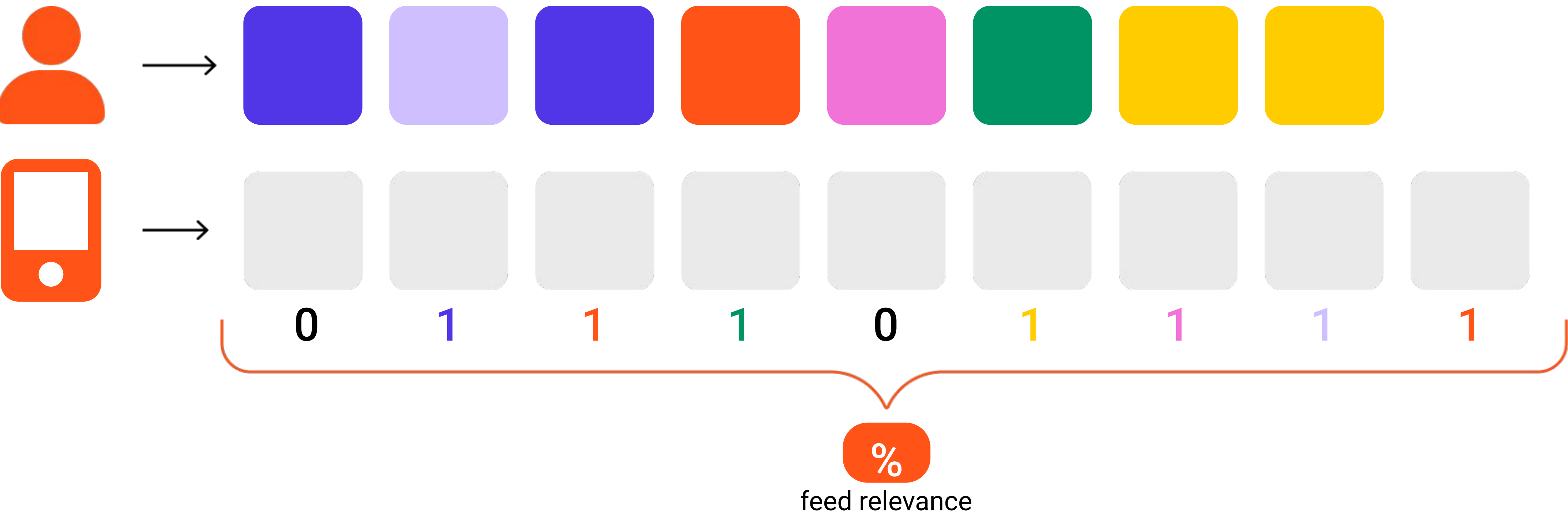
4

Дзен

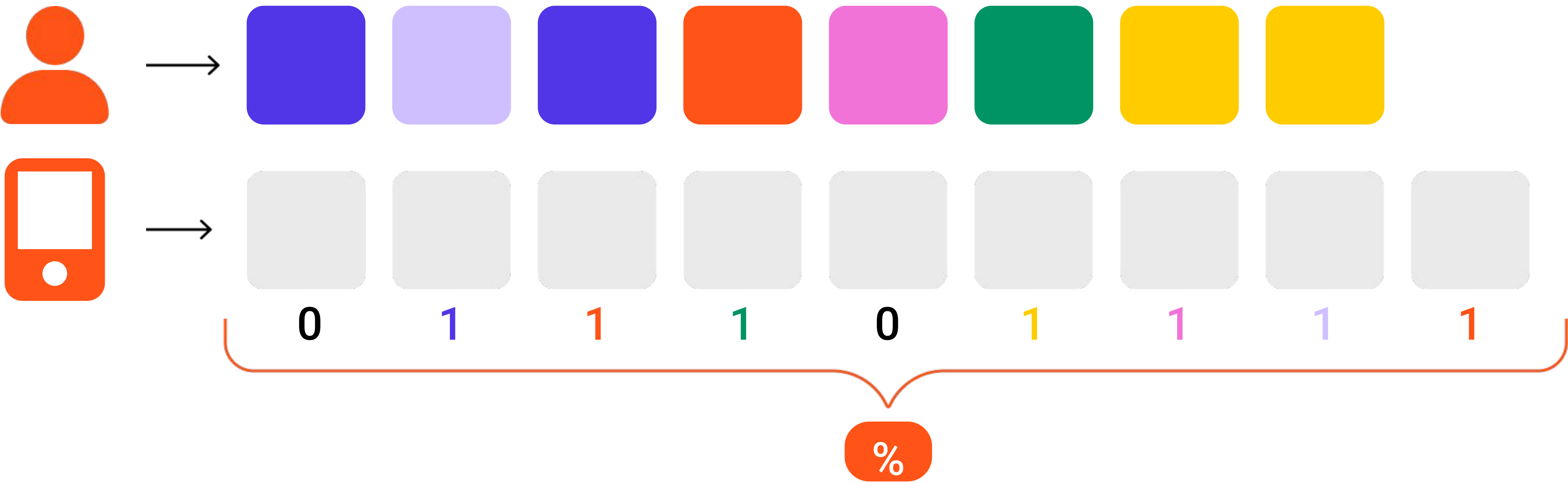
Saint HighLoad++
HL 2023

Problems

✦ Some items are not similar to user history



TODO insert results

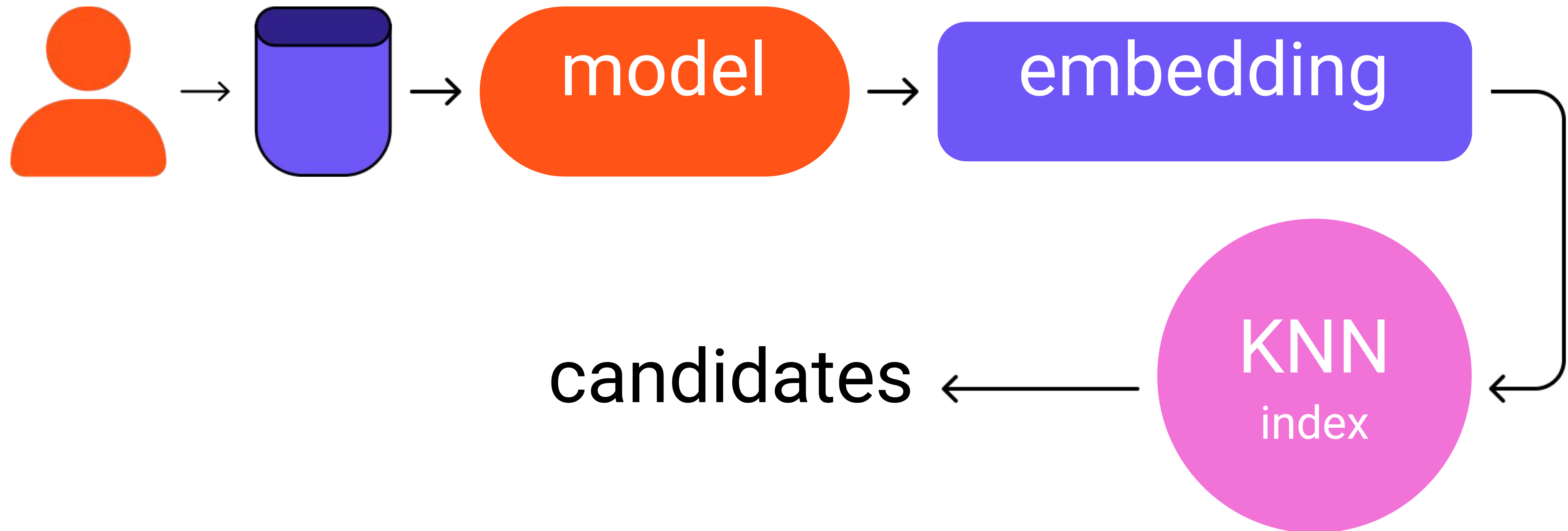


What do we want?

- 1 Make candidate generation simple
- 2 90% of shows must be relevant to user history

Problems

Not all interests are presented in feed



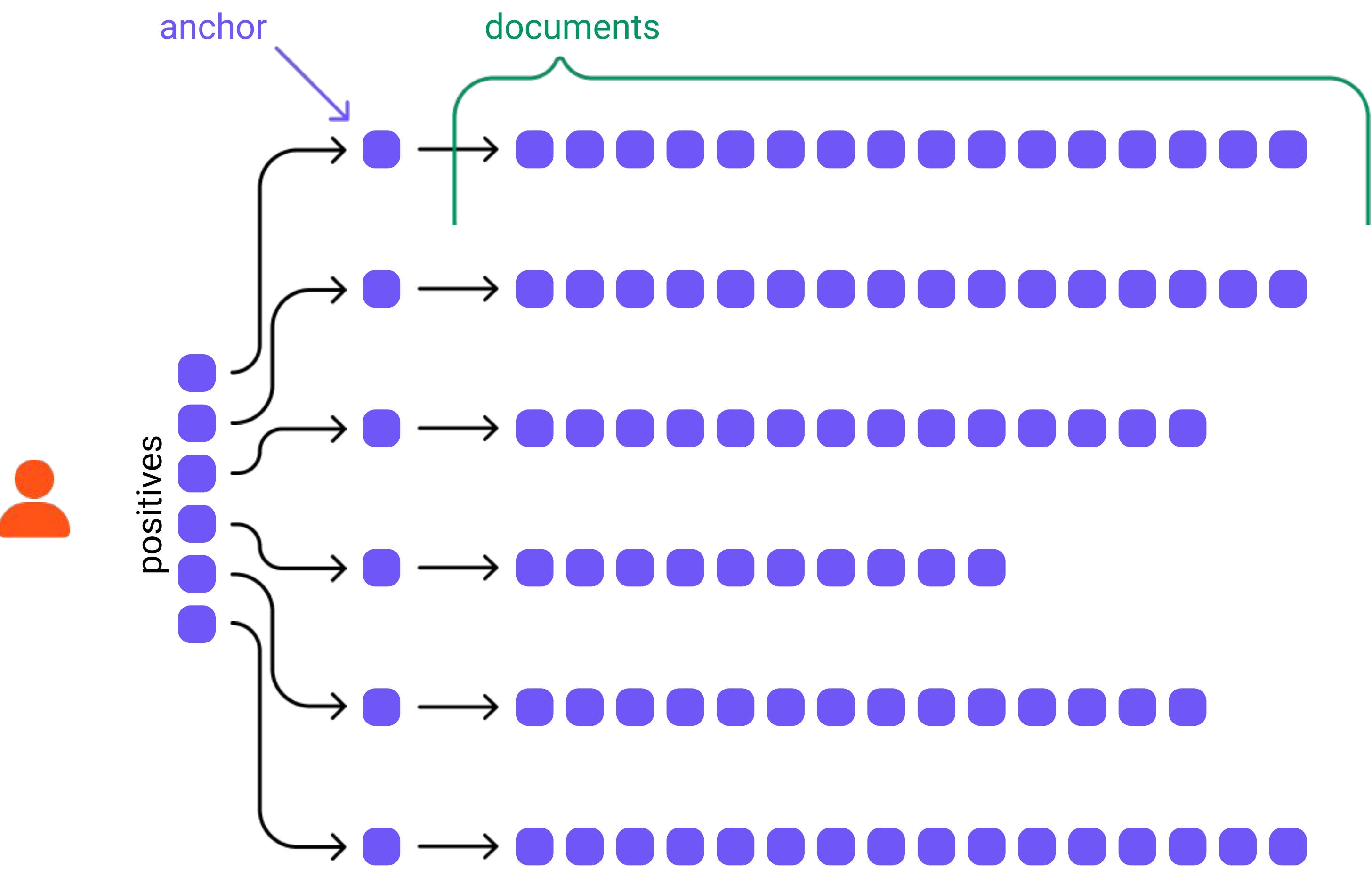
What do we want?

- 1 Make candidate generation simple
- 2 90% of shows must be relevant to user history
- 3 Guarantee usage of the whole history

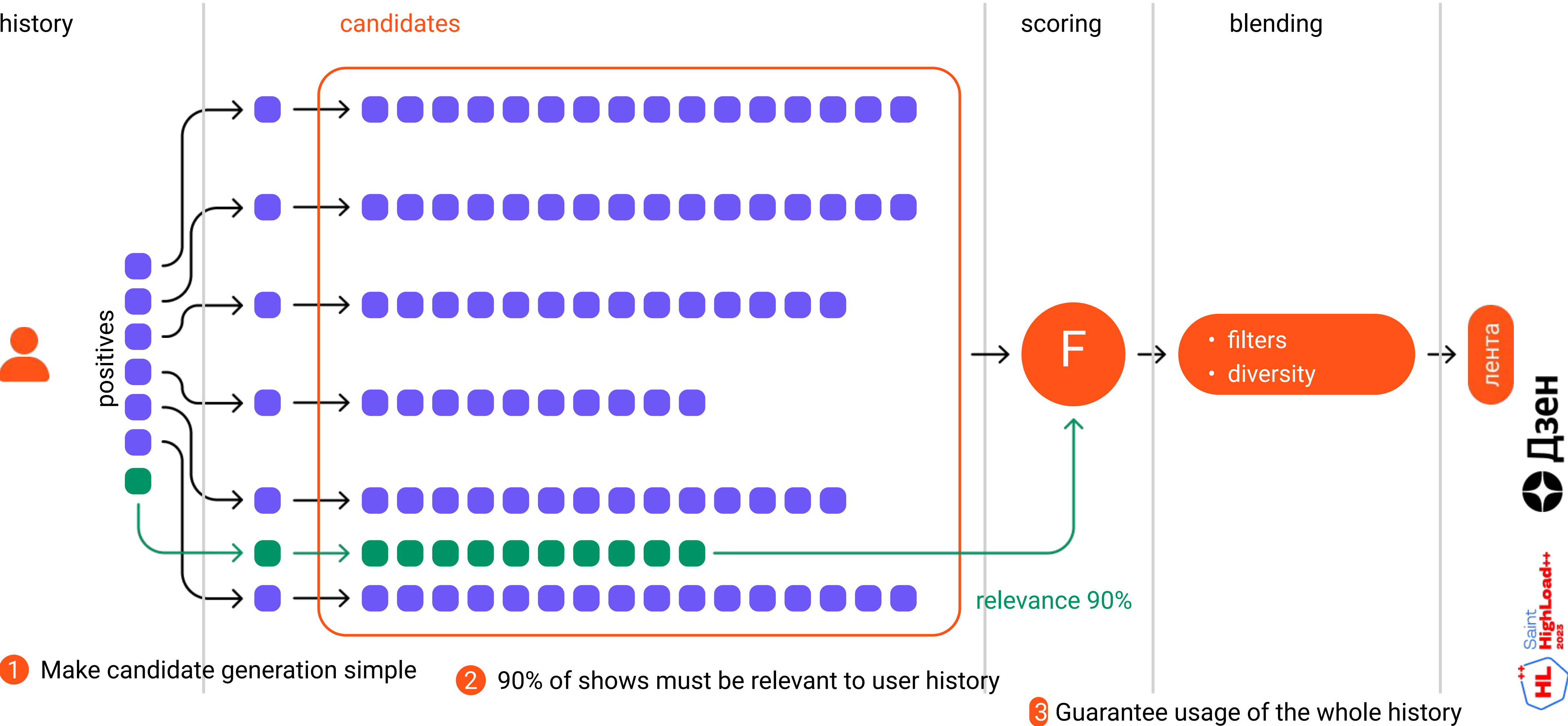
item2item approach



Item2Item



Item2Item



Item2Item: how to form a list

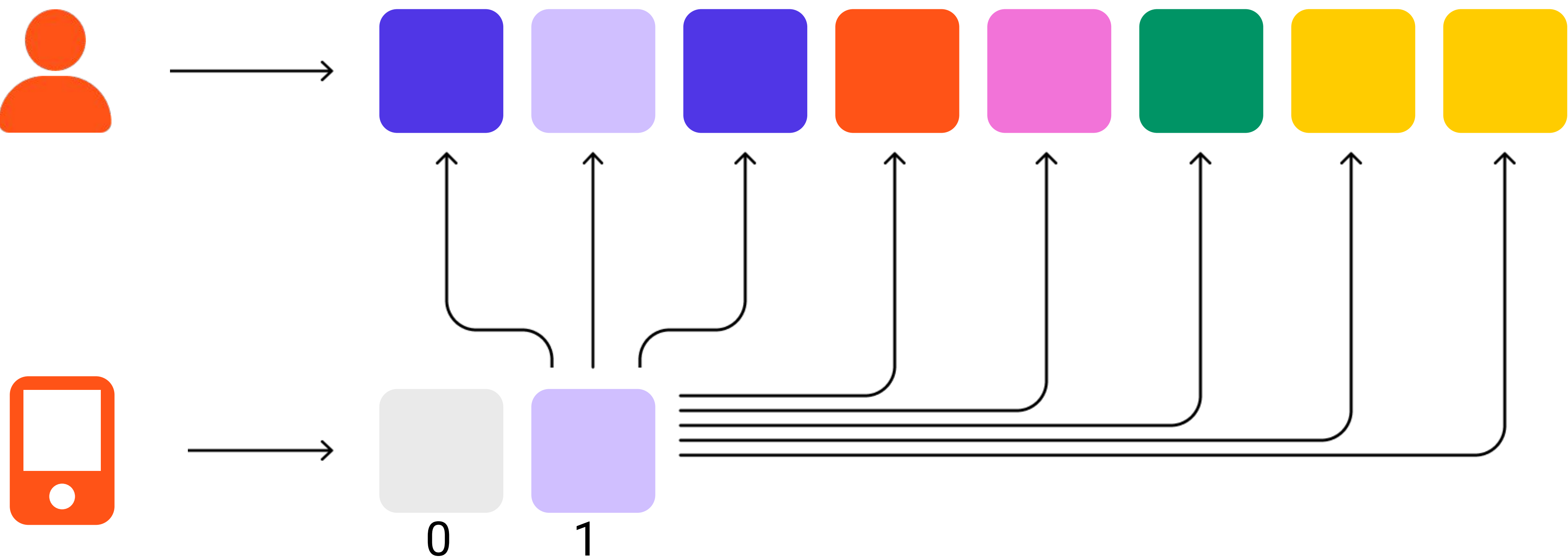
Retrieve list of 90% of relevant items for each anchor

Solution

- ✦ relevance classifier $f(A, \text{item})$
- ✦ target: ?

Problems

Part of recommended items is not similar to user history



Item2Item: how to form a list

Retrieve list of 90% of relevant items for each anchor

Solution

- ✦ relevance classifier $f(A, \text{item})$
- ✦ target: similarity by crowd labeling

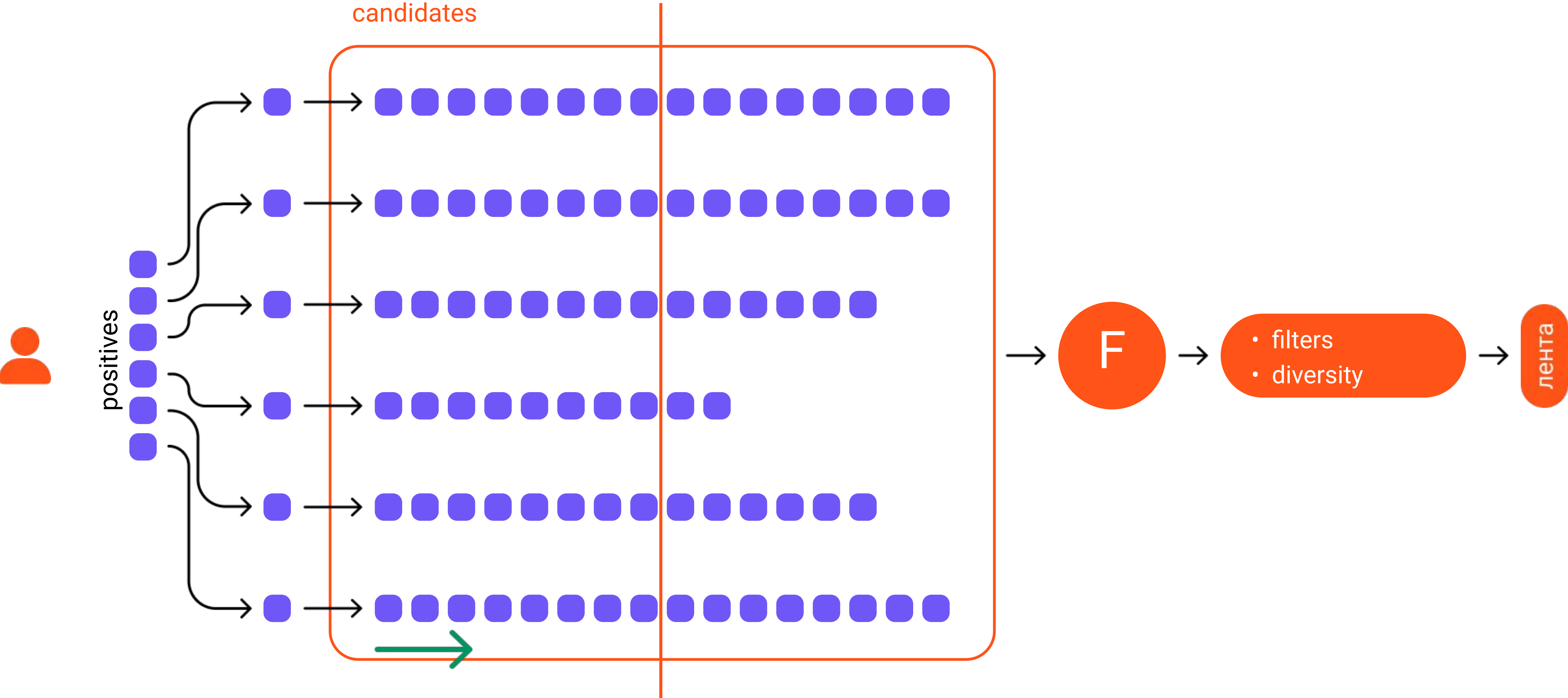
Item2Item: how to form a list

Retrieve list of 90% of relevant items for each anchor

Solution

- ✦ relevance classifier $f(A, \text{item})$
- ✦ target: similarity by crowd labeling
- ✦ features:
 - ✦ content similarity
 - ✦ authors similarity
 - ✦ documents metainformation
- ✦ to build a list we score all content base

Item2Item



Item2Item: how to build a list

Items should be ranked by attractiveness

Solution 1

✦ rank by «conditional ctr»

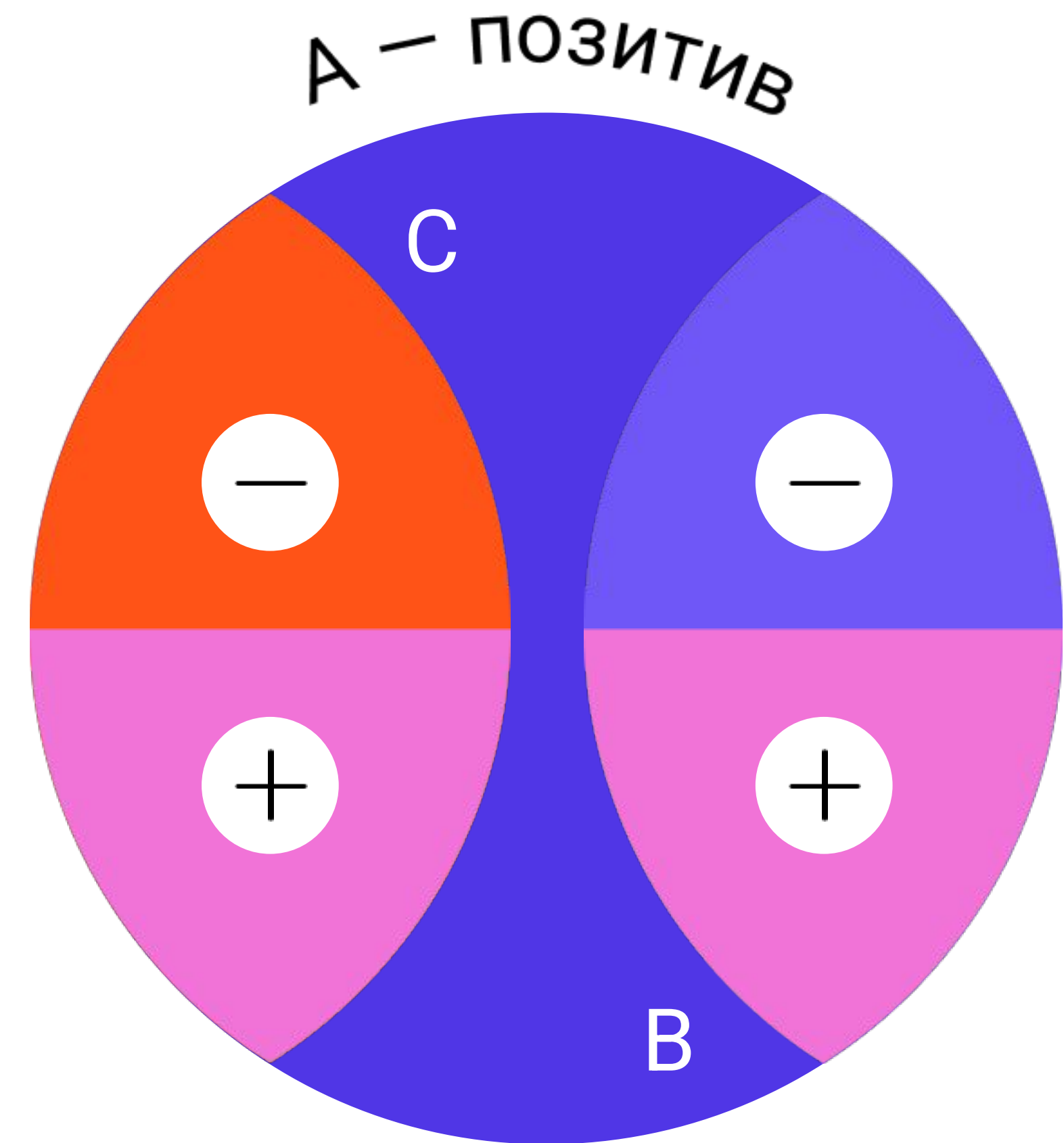
Pros

✦ it works

Cons

✦ data is sparse

✦ fails on cold items

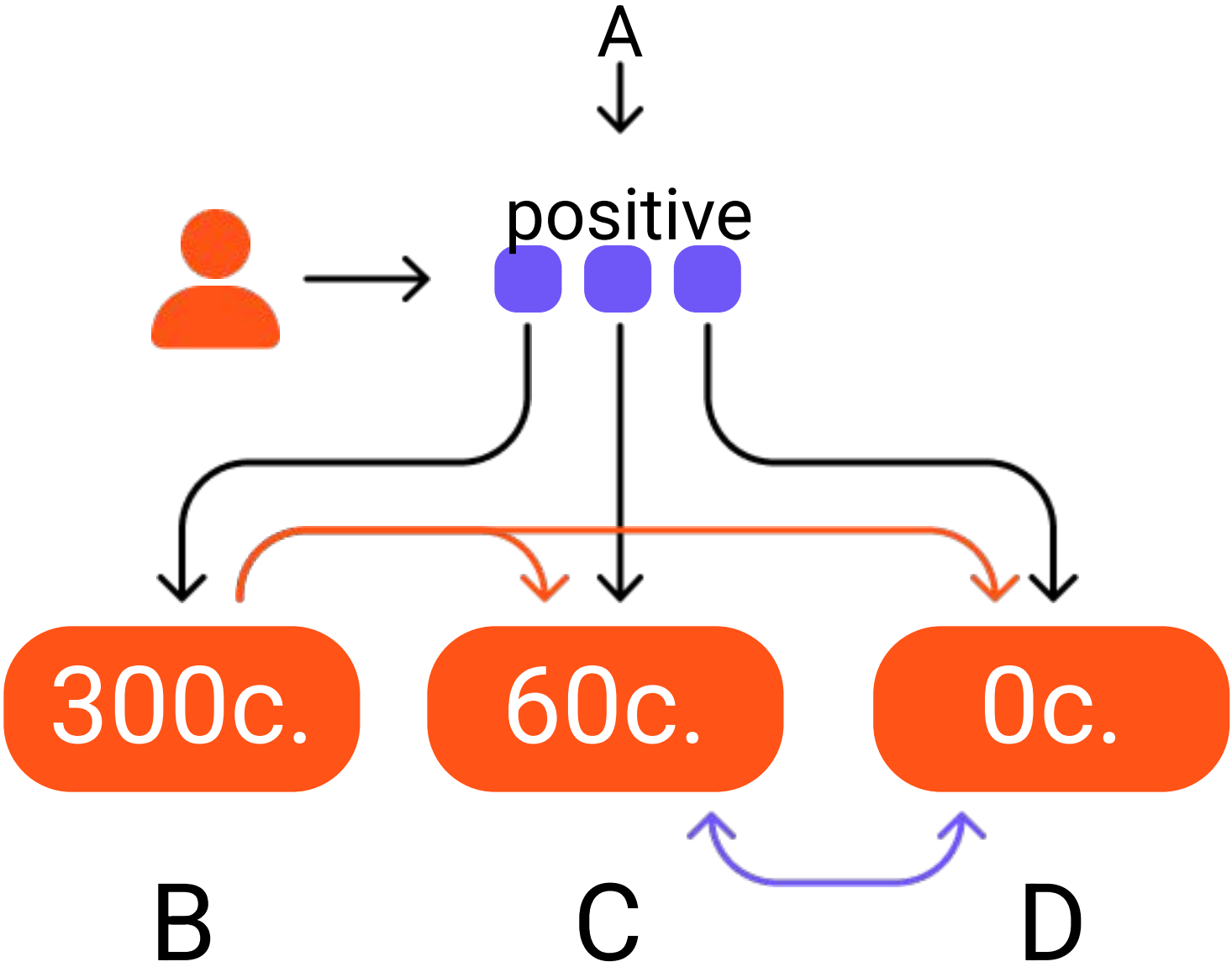


Item2Item: how to build a list

Items should be ranked by attractiveness

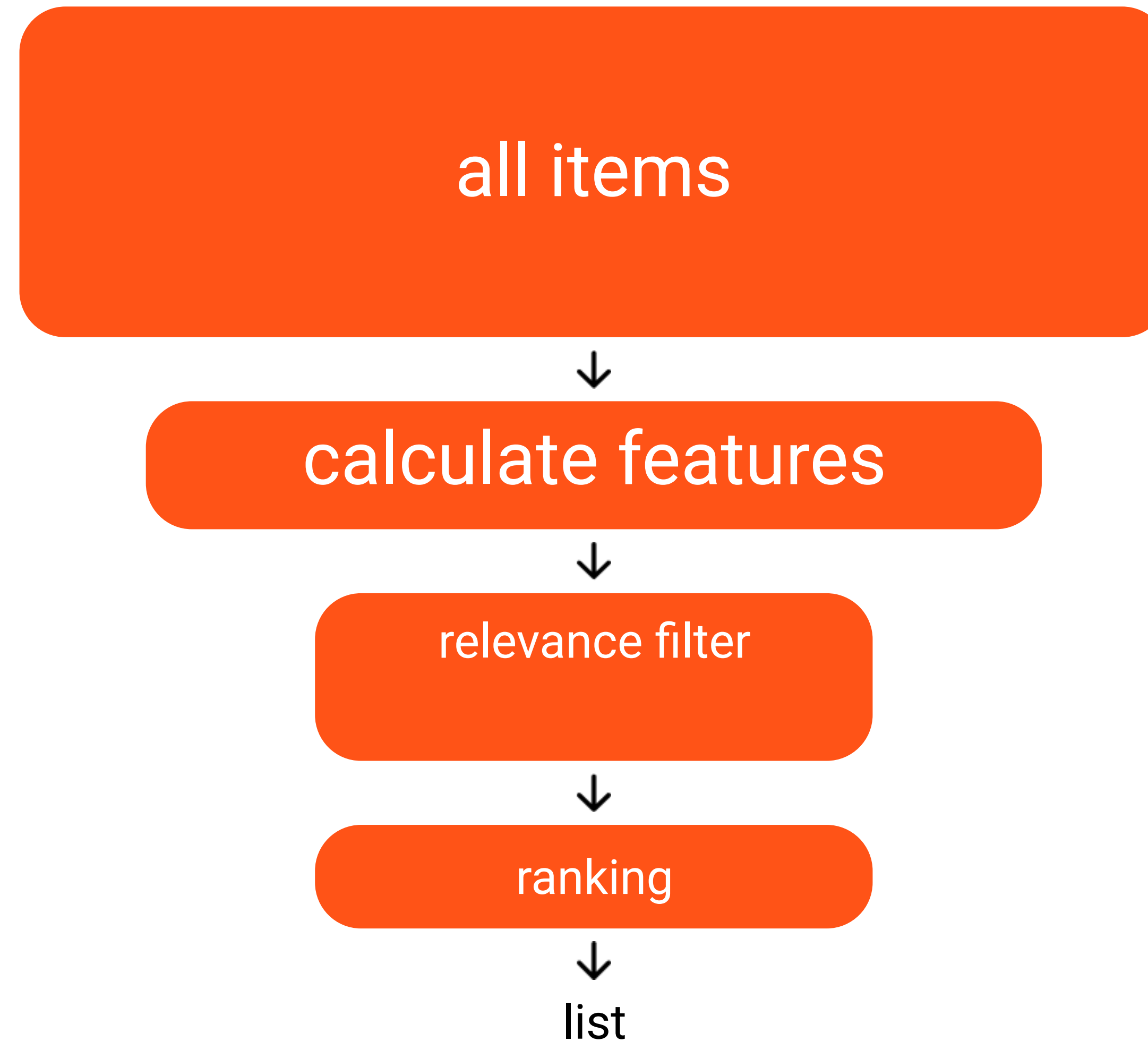
Solution 2

- ✦ log positive
- ✦ collect pairwise dataset
- ✦ train CatBoost
- ✦ features
 - ✦ collaborative similarities
 - ✦ content similarities
 - ✦ publisher similarities

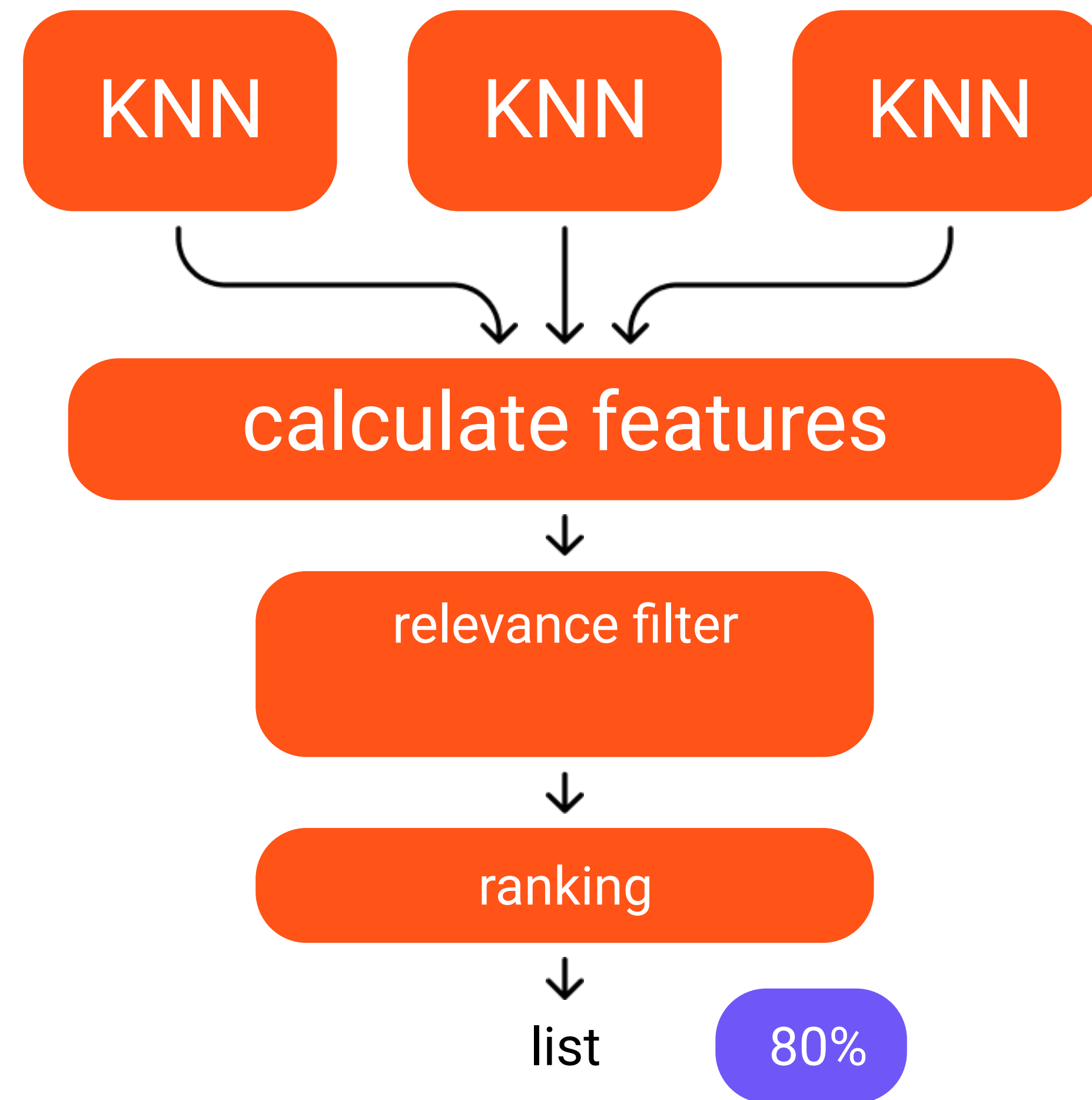


positive	winner	loser	weight
A	B	C	300—60
A	B	D	300
A	C	D	60

Item2Item: how to build a list



Item2Item: how to build a list



Item2Item: how to build a list

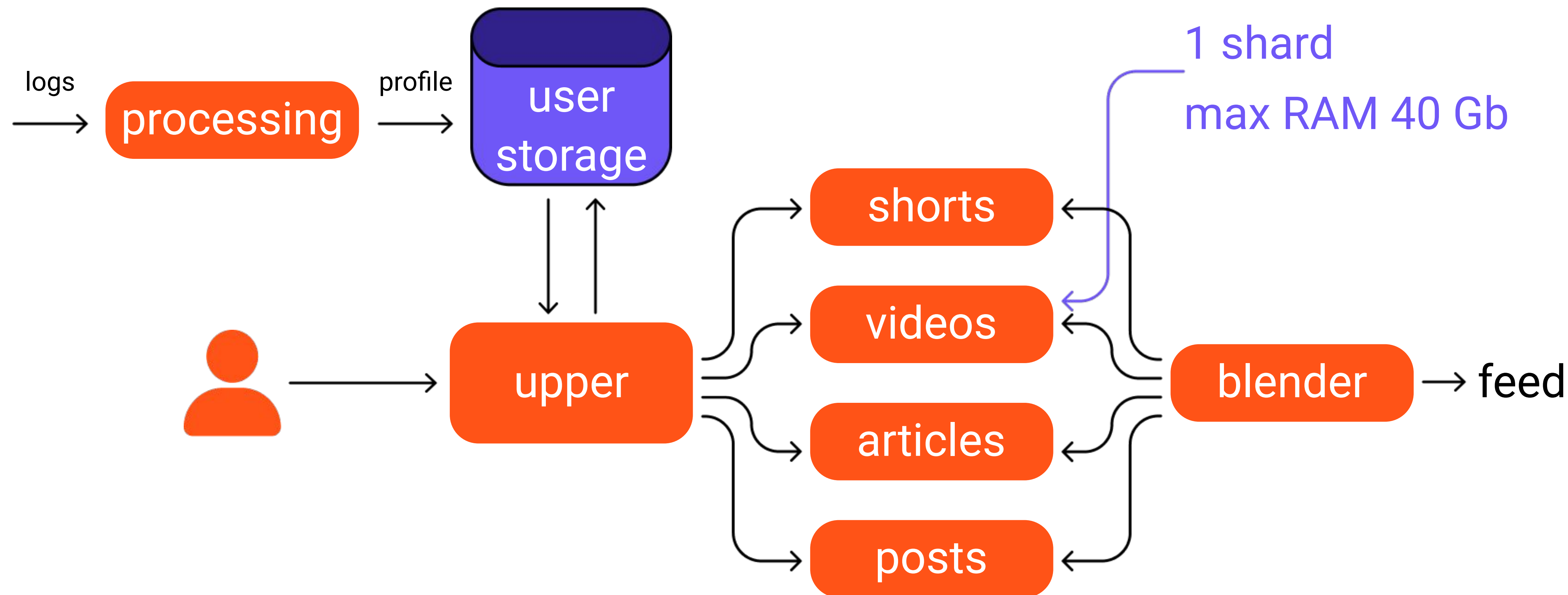
Final lists:

- ✦ 40 mln positives
- ✦ 5 mln documents
- ✦ max length 500
- ✦ 2 hours
- ✦ ~200 GB

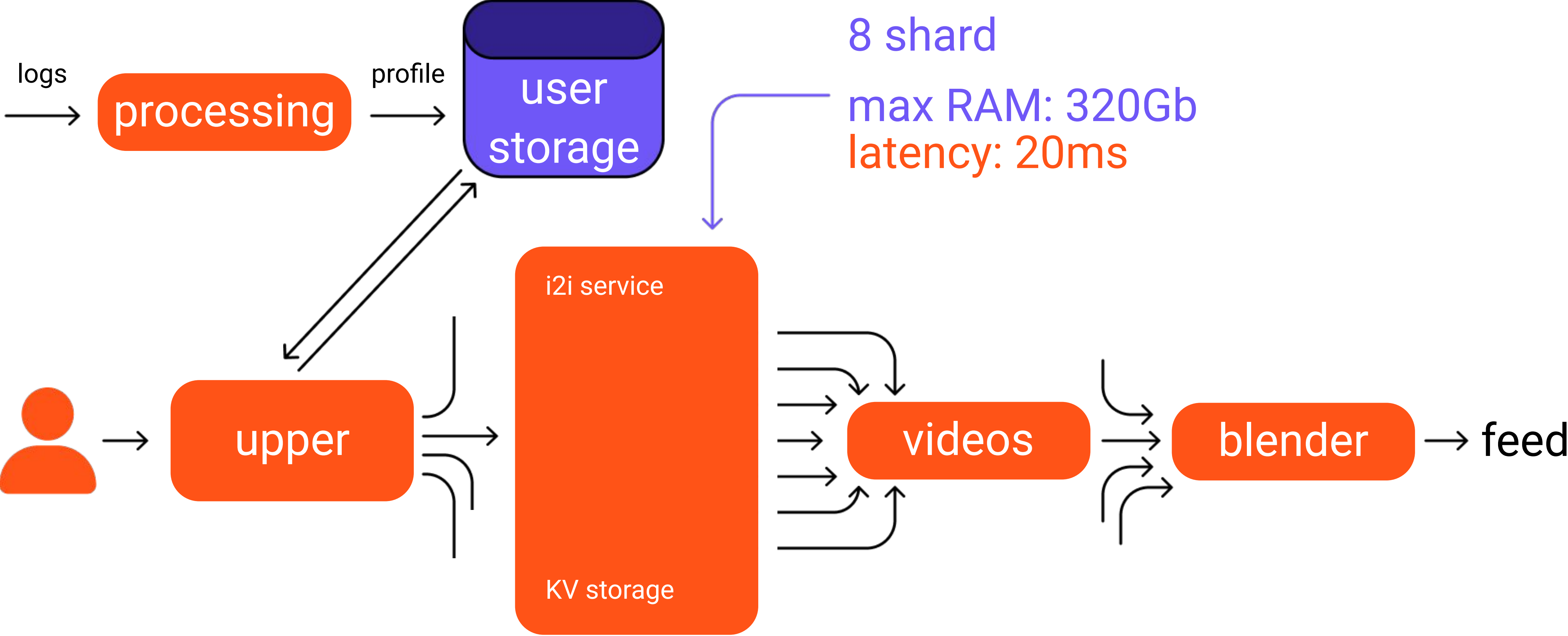
item2item service



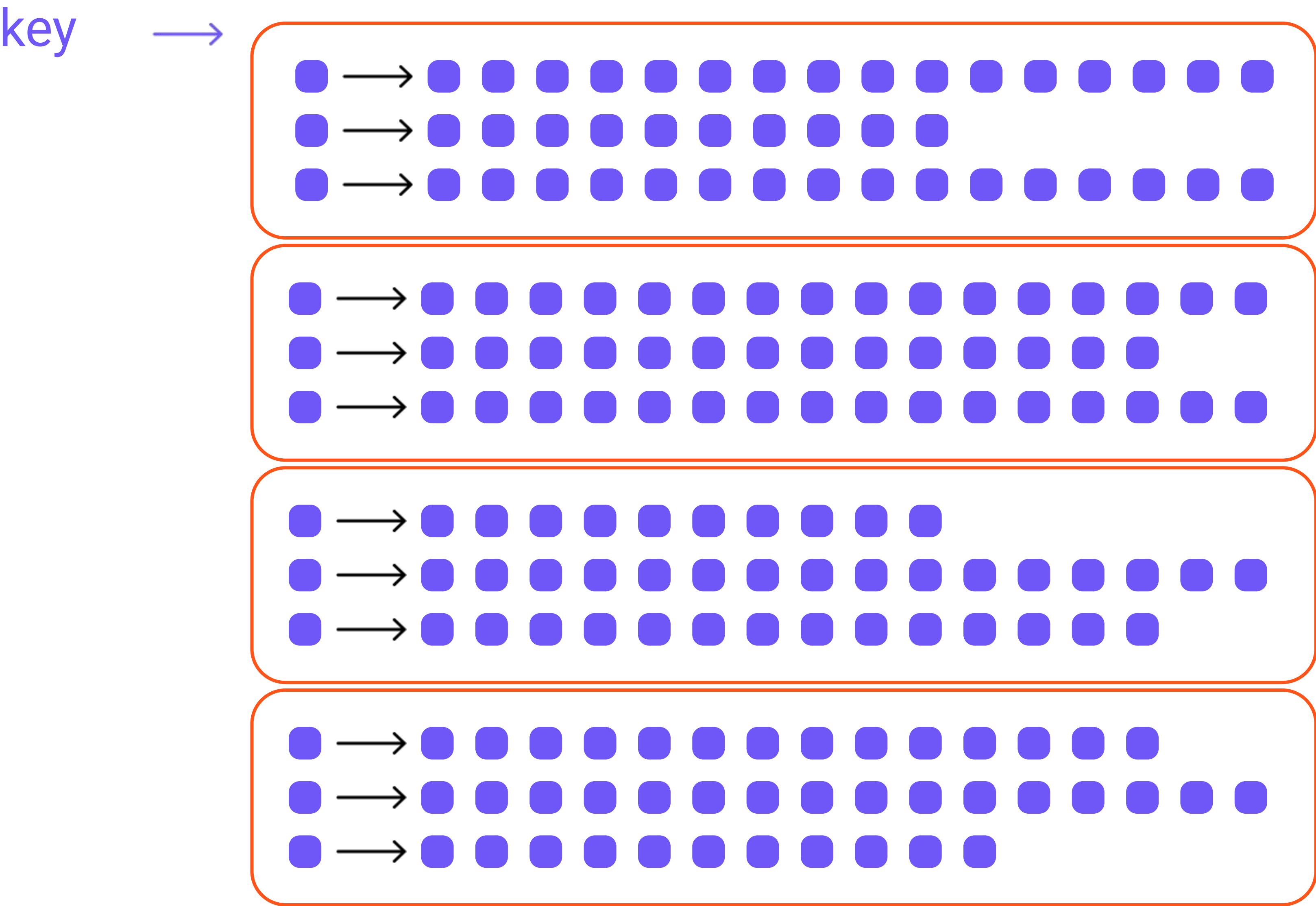
Recommendations runtime at Dzen



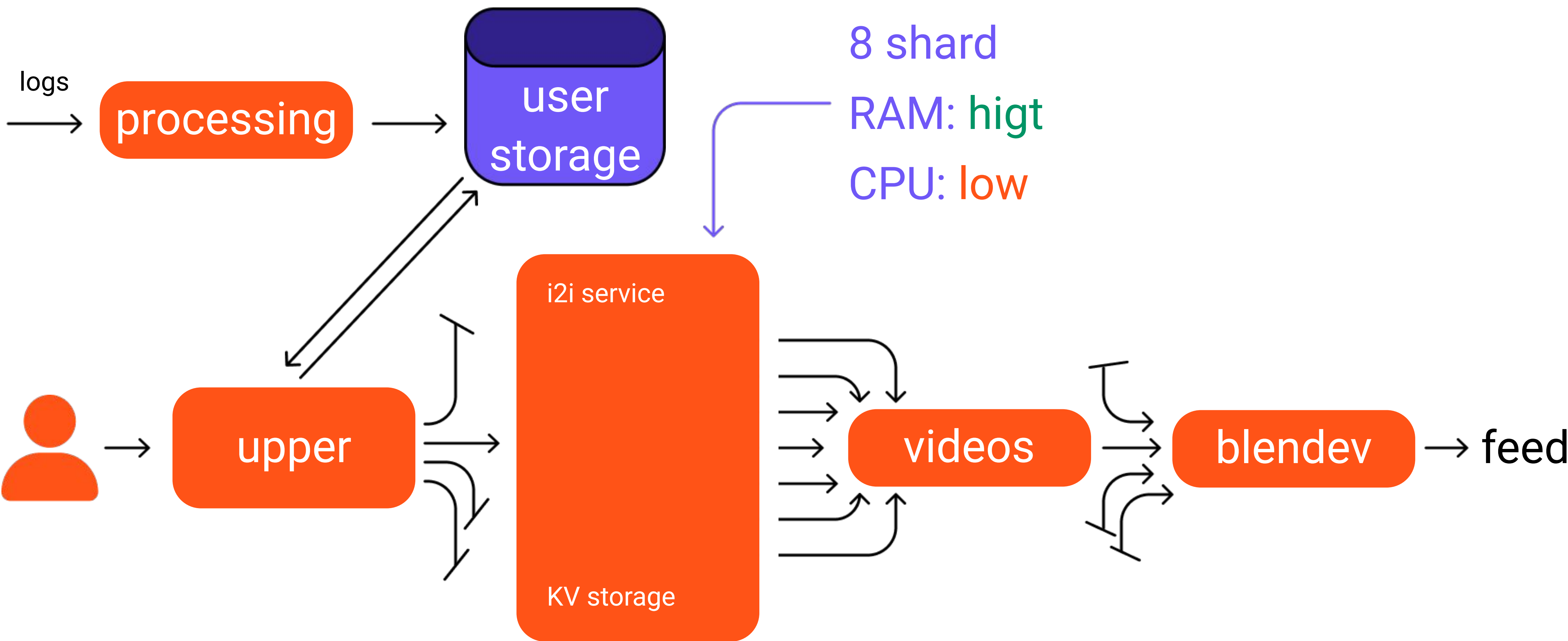
Recommendations runtime at Dzen



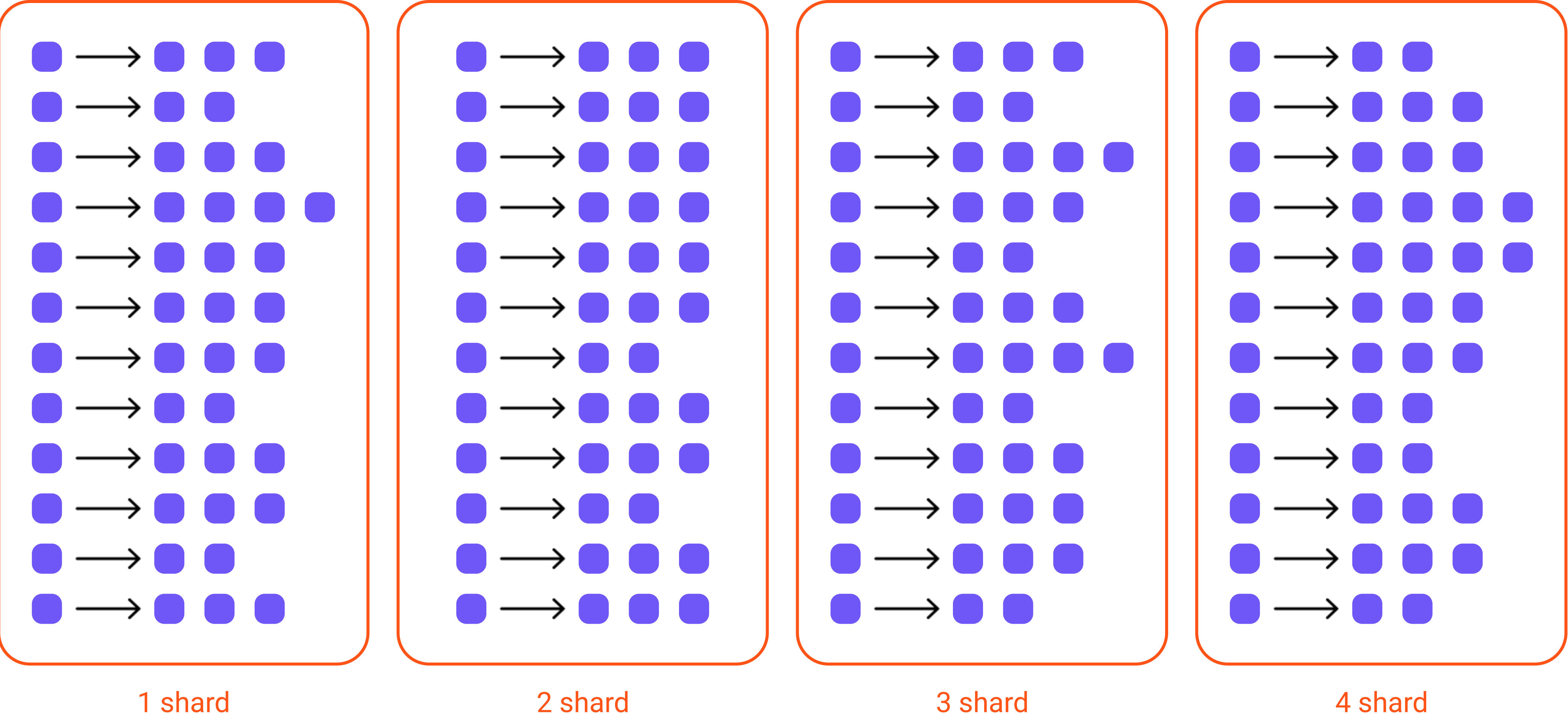
Item2Item service



Item2Item service



Item2Item service

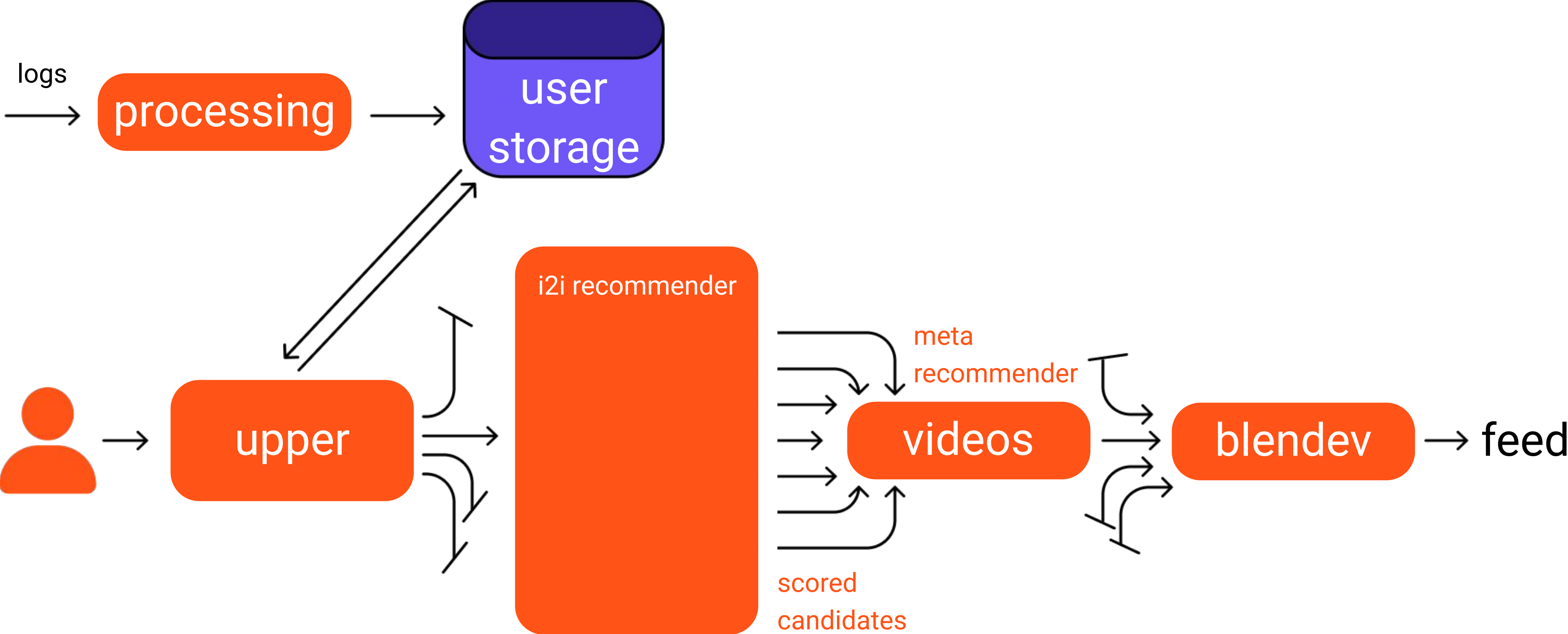


Item2Item service

1 shard:

- ✦ 1/8 documents
- ✦ what if add embeddings?

Item2Item recommender



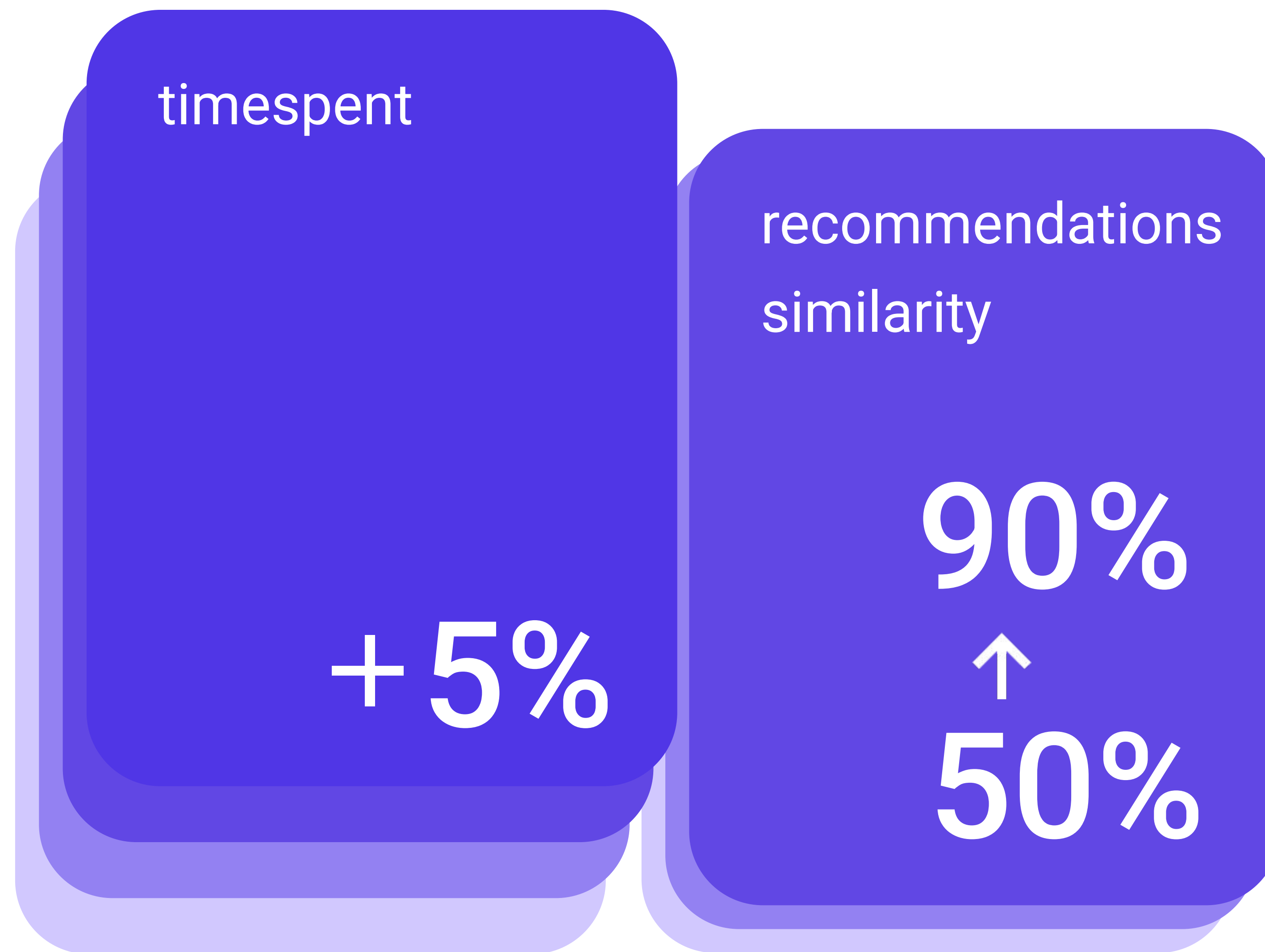
Item2Item recommender

- ✦ easy to shard
- ✦ score x8 items
- ✦ pipeline response time decreased!
- ✦ used more hardware

Results

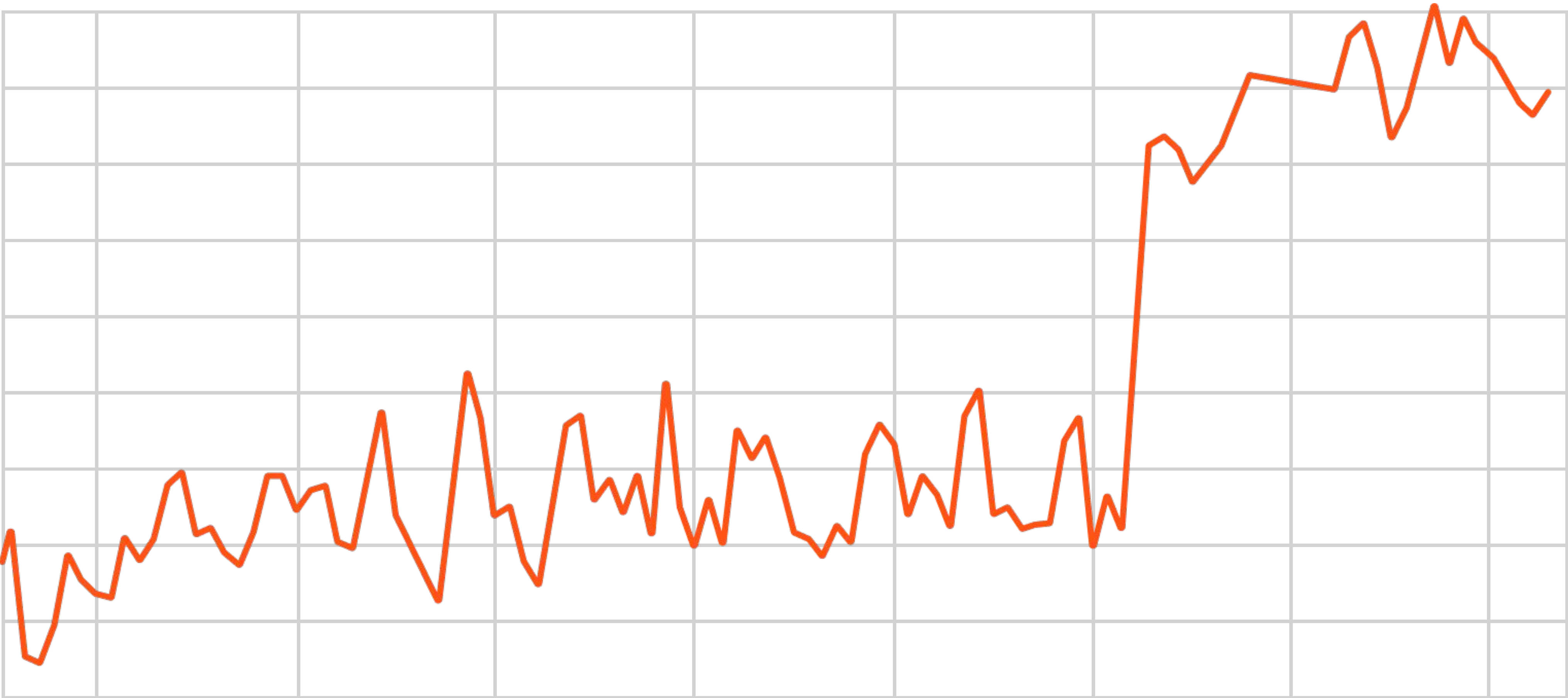


Results



NPS

Average recommendations score



Results

Recommender became simple and straightforward

- ✦ easy to debug
- ✦ more telemetry

12 HNSW

1 i2i list

Results

- ✦ recommender became simpler and clearer
- ✦ recommendations are explainable
- ✦ guarantee usage of the whole history in candidate generation process
- ✦ controllable balance between exploration and exploitation

